

Course 5 – Intelligent Application Onboarding

Module 1 – Introduction to Application Onboarding

What is Application Onboarding?

Definition

Application Onboarding is the process of integrating an application into Saviynt Identity Cloud so that Saviynt can discover, manage, govern, provision, and monitor user access to that application.

After onboarding, Saviynt can perform Identity Governance functions such as:

- Import Users
- Import Accounts
- Import Entitlements
- Access Requests
- Provisioning
- Deprovisioning
- Certifications
- Compliance Reporting
- Audit Reporting

Simple Definition

Application Onboarding is the process of adding an application to Saviynt so that user identities and access can be managed centrally.

Why is Application Onboarding Required?

Organizations use hundreds of applications.

Examples:

- Active Directory
- SAP
- Salesforce
- AWS
- Oracle
- ServiceNow
- Microsoft 365
- Workday

Without onboarding these applications into Saviynt:

- User access cannot be managed centrally.
- Access requests cannot be automated.
- Accounts cannot be provisioned.
- Compliance becomes difficult.
- Audit reporting becomes incomplete.

Application onboarding brings all applications under a single Identity Governance platform.

Why Do Organizations Onboard Applications?

Organizations onboard applications to:

- Centralize identity management.
- Automate account provisioning.
- Automate account deprovisioning.
- Improve security.
- Meet compliance requirements.
- Reduce manual effort.
- Standardize governance policies.
- Increase visibility into application access.
- Support business growth.

Real Example

A company has:

- Active Directory
- SAP
- Salesforce
- AWS
- ServiceNow

Without Saviynt:

Each application manages users independently.

IT teams manually create and remove accounts.

With Saviynt:

All applications are onboarded.

One access request in Saviynt automatically provisions the required application accounts.

Identity Governance and Application Onboarding

Identity Governance ensures that:

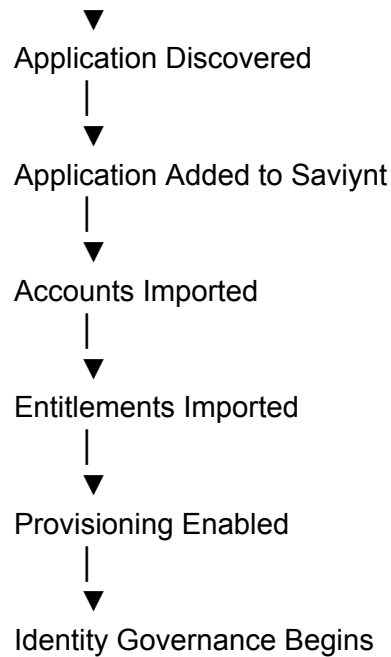
- The right user receives the right access.
- Access is approved before being granted.
- User activities are auditable.
- Access is removed when no longer needed.

Application Onboarding is the first step in this process because Saviynt must know about an application before it can govern it.

Relationship

Application Onboarding

|



Types of Applications

Saviynt supports three onboarding models:

1. Connected Applications
 2. Disconnected Applications
 3. Hybrid Applications
-

Connected Applications

Definition

A Connected Application is an application that has a **direct integration** with Saviynt using a connector.

Saviynt can automatically:

- Import accounts.
- Import entitlements.

- Provision users.
 - Deprovision users.
 - Perform reconciliation.
-

Examples

- Active Directory
 - Azure AD
 - AWS
 - Salesforce
 - SAP
 - Oracle
 - Workday
-

Advantages

- Fully automated.
 - Faster onboarding.
 - Automatic provisioning.
 - Automatic reconciliation.
 - Lower manual effort.
 - Better compliance.
-

Workflow

Saviynt



Connector



Target Application



Automatic Provisioning



Automatic Reconciliation

Disconnected Applications

Definition

A Disconnected Application is an application that **does not have a direct integration** with Saviynt.

Provisioning and reconciliation are performed manually or by using files such as CSV.

Examples

- Legacy HR systems
 - Homegrown applications
 - Custom ERP systems
 - Older applications without APIs
-

Characteristics

- Manual account import.
 - Manual provisioning.
 - CSV imports.
 - Flat file imports.
 - Administrator intervention required.
-

Workflow

Target Application



CSV File



Saviynt



Manual Provisioning

Hybrid Applications

Definition

A Hybrid Application combines automated reconciliation with manual provisioning.

Saviynt imports users and accounts automatically, but provisioning is completed through an external ticketing system such as ServiceNow.

How It Works

1. User requests access in Saviynt.
 2. Manager approves the request.
 3. Saviynt creates a provisioning task.
 4. A ticket is created in ServiceNow.
 5. The application administrator completes the task manually.
 6. Saviynt receives the completion status.
-

Advantages

- Supports applications without direct provisioning.
- Keeps approval workflows inside Saviynt.

- Maintains governance and auditability.
 - Integrates with existing ITSM processes.
-

Provisioning

Definition

Provisioning is the process of **creating, updating, or enabling user accounts and access in a target application.**

Provisioning occurs **after an access request is approved.**

Examples of Provisioning

- Create Active Directory account.
 - Create SAP account.
 - Assign AWS IAM Role.
 - Add a user to an AD Group.
 - Assign a Salesforce Role.
-

Provisioning Flow

Access Request

↓

Approval

↓

Saviynt

↓

Provisioning



Target Application

Deprovisioning

Definition

Deprovisioning is the process of **removing, disabling, or revoking user access** from an application.

It is usually triggered when:

- Employee resigns.
 - Contractor leaves.
 - Access is revoked.
 - Certification removes access.
-

Examples

- Disable Active Directory account.
 - Remove SAP role.
 - Remove AWS IAM permissions.
 - Delete Salesforce account.
-

Why is Deprovisioning Important?

Without deprovisioning:

- Former employees may still access company systems.
 - Security risks increase.
 - Compliance requirements are violated.
-

Reconciliation

Definition

Reconciliation is the process of **importing the latest accounts, users, and entitlements from a target application into Saviynt**.

Direction:

Target Application



Saviynt

Its purpose is to keep Saviynt synchronized with the actual state of the application.

Various Reconciliation Use Cases

Saviynt performs reconciliation for many purposes.

Import Users

Import users from the application into Saviynt.

Import Accounts

Import application accounts and associate them with identities.

Import Entitlements

Import:

- Groups
- Roles

- Permissions

into Saviynt.

Enable or Disable Accounts

Detect whether accounts have been enabled or disabled directly in the target application.

Password Synchronization

Import password status (where supported) or synchronize password-related information.

Add or Remove Access

Detect changes made directly in the target application and synchronize them with Saviynt.

Various Provisioning Use Cases

Saviynt supports many provisioning operations.

Create Account

Create a new application account.

Update User

Update user attributes.

Examples:

- Department
- Email
- Manager

Update or Remove Accounts

Modify or delete application accounts.

Enable or Disable Accounts

Activate or deactivate user accounts.

Change Password

Reset or change passwords where supported.

Create Service Desk Ticket

For hybrid or disconnected applications, create a ServiceNow ticket for manual provisioning.

Configure Provisioning Tasks

Generate tasks for administrators to complete manually.

Fetch Ticket Status

Retrieve the status of ServiceNow or ITSM tickets and update Saviynt.

Add or Remove Access

Assign or revoke groups, roles, or permissions.

Emergency Access Provisioning

Grant temporary emergency access IDs for urgent situations and revoke them afterward.

Application Onboarding Process

Saviynt simplifies onboarding into two main phases.

Step 1 – Discover

Identify applications within the organization.

Discovery methods include:

- ServiceNow CMDB
- File Upload (CSV/Excel)

The goal is to build a complete inventory of applications.

Step 2 – Onboard

After discovery, applications are added to Saviynt.

During onboarding:

- The application is configured.
 - Connectors are assigned (if available).
 - Endpoints are created.
 - Governance policies are applied.
 - Provisioning and reconciliation are enabled as required.
-

Complete Workflow

Identify Applications



Discover Applications



Review Inventory



Onboard Application



Configure Connector



Import Accounts



Import Entitlements



Provision Users



Govern Access

Benefits of Application Onboarding

Application Onboarding provides many advantages.

Centralized Governance

All applications are managed from one platform.

Better Visibility

Administrators know:

- Which applications exist.
 - Who has access.
 - What permissions users have.
-

Automated Provisioning

Accounts are created automatically.

Automated Deprovisioning

Access is removed automatically when it is no longer needed.

Improved Compliance

Supports:

- SOX
- HIPAA
- GDPR
- ISO 27001
- PCI DSS

by providing complete audit trails.

Reduced Manual Effort

Less manual account creation and fewer human errors.

Scalability

Organizations can onboard hundreds of applications using a consistent process.

Real Project Example

ABC Bank uses the following applications:

- Active Directory
- SAP
- AWS
- Salesforce
- ServiceNow

Initially, each application is managed separately.

The bank implements Saviynt.

Step 1 – Discover

Saviynt discovers all applications using ServiceNow CMDB.

Step 2 – Onboard

Each application is onboarded into Saviynt.

For connected applications:

- Connectors are configured.
- Accounts and entitlements are imported.
- Provisioning is enabled.

For disconnected applications:

- CSV imports are configured.
 - Manual provisioning tasks are created.
-

Result

A new employee joins the Finance department.

HR creates the employee record.

Saviynt provisions:

- Active Directory account.
- SAP account.
- Microsoft 365 mailbox.
- AWS access.

If the employee later resigns:

- All accounts are automatically disabled or removed.
 - Access is revoked.
 - Audit logs are updated.
-

Best Practices

- Discover all enterprise applications before onboarding.
 - Prefer connected applications whenever possible.
 - Use standard connectors instead of custom integrations when available.
 - Onboard only applications that require identity governance.
 - Regularly reconcile accounts and entitlements.
 - Test onboarding in a non-production environment before moving to production.
 - Maintain consistent naming conventions for applications, endpoints, and security systems.
 - Monitor provisioning and reconciliation jobs regularly.
-

Interview Questions

Q1. What is Application Onboarding?

Application Onboarding is the process of integrating an application into Saviynt so that identities, accounts, and access can be governed centrally.

Q2. Why is Application Onboarding important?

It enables centralized identity governance, automated provisioning, compliance, audit reporting, and better visibility into application access.

Q3. What are the three types of applications in Saviynt?

Connected, Disconnected, and Hybrid applications.

Q4. What is the difference between Provisioning and Reconciliation?

Provisioning sends changes from **Saviynt to the target application**, while Reconciliation imports data from the **target application into Saviynt**.

Q5. What is a Hybrid Application?

A Hybrid Application uses automated reconciliation but manual provisioning, often through a ticketing system such as ServiceNow.

Q6. What are common reconciliation use cases?

Importing users, accounts, entitlements, account status changes, and detecting access changes made directly in target applications.

Q7. What are common provisioning use cases?

Creating accounts, updating users, assigning or removing access, enabling/disabling accounts, changing passwords, and creating provisioning tasks or ServiceNow tickets.

Quick Revision

- Application Onboarding integrates applications into Saviynt for centralized governance.
- Organizations onboard applications to automate identity and access management.
- Connected applications support automated provisioning and reconciliation.
- Disconnected applications rely on manual processes or file imports.
- Hybrid applications combine automated reconciliation with manual provisioning.
- Provisioning creates or updates accounts in target systems.
- Deprovisioning removes or disables access.
- Reconciliation synchronizes account and entitlement data from target applications to Saviynt.
- Application Onboarding follows two main phases: **Discover** and **Onboard**.
- Proper onboarding improves security, compliance, visibility, and operational efficiency.

Module 2 – Application Discovery

What is Application Discovery?

Definition

Application Discovery is the process of identifying all the applications used within an organization and bringing their information into Saviynt before they are onboarded.

Application Discovery helps organizations understand:

- What applications exist
- Who owns them
- Which department uses them
- Whether they are already governed
- Whether they need to be onboarded into Saviynt

It is the **first step** in the Application Onboarding process.

Simple Definition

Application Discovery is the process of finding and collecting information about applications that need to be managed by Saviynt.

Why is Application Discovery Important?

Large organizations may use hundreds or even thousands of applications.

Examples:

- SAP
- Salesforce
- Active Directory
- AWS

- Oracle
- Workday
- ServiceNow
- Microsoft 365

Without discovery:

- Some applications may be forgotten.
- User access cannot be governed.
- Shadow IT applications may remain unmanaged.
- Compliance reports become incomplete.

Application Discovery ensures every important application is identified before onboarding.

Objectives of Application Discovery

Application Discovery helps organizations:

- Identify all business applications.
 - Build an application inventory.
 - Eliminate duplicate application records.
 - Identify unmanaged applications.
 - Plan onboarding priorities.
 - Improve governance and compliance.
-

Discovery Sources

Definition

A **Discovery Source** is the location or system from which Saviynt collects application information.

Saviynt supports multiple discovery sources.

Common discovery sources include:

- ServiceNow CMDB
- CSV or Excel File Upload

- Internal Application Database
 - Custom Discovery Integrations
-

ServiceNow CMDB Discovery

What is CMDB?

CMDB stands for **Configuration Management Database**.

It is a database maintained in ServiceNow that stores information about an organization's IT assets.

These assets include:

- Applications
 - Servers
 - Databases
 - Network Devices
 - Cloud Resources
-

What is ServiceNow CMDB Discovery?

Saviynt connects to the ServiceNow CMDB and imports application information automatically.

Instead of manually entering application details, Saviynt retrieves them directly from the CMDB.

Information Imported from CMDB

Typical information includes:

- Application Name
- Application Owner
- Business Unit
- Environment (Development, Test, Production)
- Criticality
- Vendor

- Description
 - Application Status
-

Why Use CMDB Discovery?

Benefits include:

- Automatic application discovery.
 - Accurate application inventory.
 - Less manual work.
 - Better synchronization with IT operations.
 - Reduced duplicate records.
-

Example

ABC Bank stores all application information in ServiceNow.

Saviynt connects to the CMDB.

Applications such as:

- SAP
- Salesforce
- AWS
- Oracle

are automatically discovered and displayed for onboarding.

File Upload Discovery

Definition

Some organizations do not maintain a CMDB.

Instead, they maintain application information in spreadsheets or CSV files.

Saviynt allows administrators to upload these files for application discovery.

Supported File Types

- CSV
 - Excel (depending on implementation)
 - Flat Files
-

Example CSV

```
ApplicationName,Owner,Department,Criticality
SAP,John Smith,Finance,High
Salesforce,Jane Doe,Sales,Medium
AWS,Ravi Kumar,IT,High
```

Saviynt imports these applications into the Discovery List.

Discovery Source

Definition

The Discovery Source identifies **where the application information originated**.

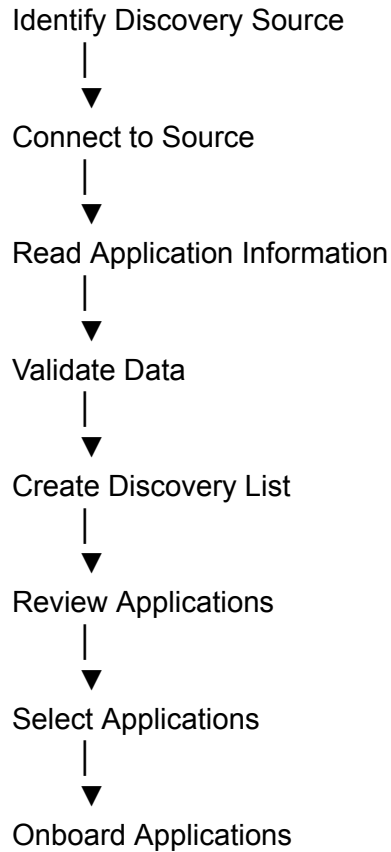
Examples:

- ServiceNow CMDB
- CSV Upload
- Internal Database
- Custom Integration

Knowing the source helps administrators validate the application's information and troubleshoot discovery issues.

Discovery Process

The Application Discovery process follows these steps:



Step 1 – Identify the Source

Choose where application information will come from.

Example:

- ServiceNow CMDB
 - CSV File
-

Step 2 – Read Application Data

Saviynt retrieves:

- Application Name
 - Owner
 - Department
 - Business Unit
 - Criticality
-

Step 3 – Validate Information

Saviynt checks:

- Duplicate application names.
 - Missing owners.
 - Missing application names.
 - Invalid metadata.
-

Step 4 – Create Discovery List

All discovered applications are displayed for administrator review.

Nothing is onboarded automatically.

Application Inventory

Definition

An **Application Inventory** is a centralized list of all discovered applications within an organization.

It acts as a catalog of business applications.

Typical Information

Each application may include:

- Application Name
 - Owner
 - Department
 - Vendor
 - Business Function
 - Criticality
 - Status
 - Environment
 - Discovery Source
-

Why is Application Inventory Important?

It helps organizations:

- Track applications.
 - Prioritize onboarding.
 - Identify unmanaged systems.
 - Improve governance.
 - Support audits.
-

Application Metadata

Definition

Metadata is descriptive information about an application.

It helps administrators understand the application's purpose and ownership.

Common Metadata

- Application Name

- Description
 - Owner
 - Department
 - Business Unit
 - Vendor
 - Environment
 - Criticality
 - Application Type
 - Discovery Date
-

Example

Application:

SAP

Metadata:

- Owner: Finance Team
 - Vendor: SAP
 - Environment: Production
 - Criticality: High
 - Department: Finance
-

Discovery List

Definition

The **Discovery List** contains all applications identified during discovery.

Applications remain in this list until an administrator decides what to do with them.

Possible Actions

- Review application.
- Hide application.

- Onboard application.
 - Ignore application.
 - Update metadata.
-

Hide Application

Definition

Sometimes discovered applications should not be onboarded.

Instead of deleting them, administrators can **hide** them.

When is Hide Used?

Examples:

- Duplicate application.
 - Test application.
 - Temporary application.
 - Obsolete application.
 - Application not managed by Saviynt.
-

Benefits

- Keeps the inventory clean.
 - Prevents accidental onboarding.
 - Preserves discovery history.
-

Onboard Application

Definition

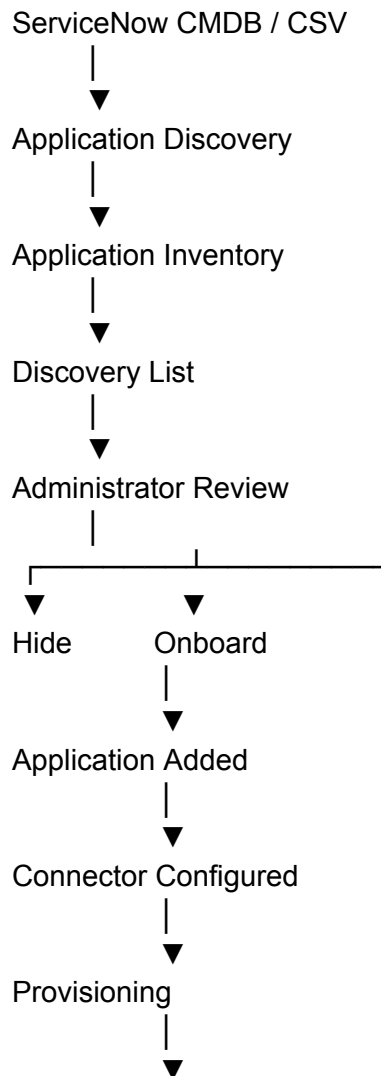
Once an application is reviewed and approved, it is onboarded into Saviynt.

During onboarding:

- Security System is created.
- Endpoint is created.
- Connector is configured (if applicable).
- Provisioning is enabled.
- Reconciliation is configured.

After onboarding, Saviynt begins governing the application.

Discovery Workflow



Real Project Example

ABC Manufacturing has more than **350 business applications**.

The company stores application information in ServiceNow CMDB.

Step 1

Saviynt connects to the CMDB.

Step 2

Saviynt discovers:

- SAP
 - Salesforce
 - Active Directory
 - AWS
 - Oracle
 - ServiceNow
-

Step 3

The applications appear in the Discovery List.

Step 4

Administrators review each application.

One outdated test application is hidden.

The remaining production applications are approved.

Step 5

Each approved application is onboarded into Saviynt.

For connected applications:

- Connectors are configured.
- Accounts and entitlements are imported.
- Provisioning and reconciliation are enabled.

For disconnected applications:

- CSV import processes and manual provisioning workflows are configured.

The organization now has a complete and governed application inventory.

Best Practices

- Discover all applications before starting onboarding.
 - Use ServiceNow CMDB whenever available to keep discovery automated and accurate.
 - Review the Discovery List before onboarding.
 - Hide duplicate or obsolete applications instead of deleting them.
 - Verify application ownership and metadata.
 - Prioritize onboarding of business-critical applications.
 - Keep the Application Inventory up to date.
 - Periodically rediscover applications to identify new systems.
-

Interview Questions

Q1. What is Application Discovery?

Application Discovery is the process of identifying applications used within an organization before onboarding them into Saviynt for identity governance.

Q2. Why is Application Discovery important?

It ensures all relevant applications are identified, inventoried, and reviewed before onboarding, improving governance, compliance, and visibility.

Q3. What are common Discovery Sources?

Common discovery sources include ServiceNow CMDB, CSV or Excel file uploads, internal application databases, and custom integrations.

Q4. What is ServiceNow CMDB Discovery?

It is the process of connecting Saviynt to the ServiceNow CMDB to automatically import application information for review and onboarding.

Q5. What is an Application Inventory?

An Application Inventory is the centralized list of all discovered applications and their metadata.

Q6. What is Application Metadata?

Application Metadata is descriptive information such as application name, owner, department, environment, vendor, and criticality.

Q7. Why would you hide an application?

Applications are hidden when they are duplicates, test systems, obsolete, temporary, or outside the scope of identity governance.

Q8. What happens when an application is onboarded?

Saviynt creates the required configuration (such as Security System, Endpoint, and Connector where applicable), imports accounts and entitlements, and enables governance processes like provisioning and reconciliation.

Quick Revision

- Application Discovery identifies applications before onboarding.
- Discovery builds a complete Application Inventory.
- Common discovery sources include ServiceNow CMDB and CSV file uploads.
- ServiceNow CMDB Discovery automates application discovery.
- Application Metadata provides descriptive information about each application.
- The Discovery List displays applications awaiting administrator review.
- Applications can be hidden if they are not required.
- Approved applications are onboarded into Saviynt.
- Application Discovery is the first step toward centralized Identity Governance.

Module 3 – Connected Application Onboarding (Part 1 & Part 2)

Part 1 – Connected Application Fundamentals

What is a Connected Application?

Definition

A **Connected Application** is an application that is directly integrated with Saviynt using a connector.

Through this integration, Saviynt can communicate automatically with the application to perform Identity Governance operations.

These operations include:

- Import Users
- Import Accounts
- Import Entitlements
- Provision Accounts
- Update Accounts
- Disable Accounts
- Remove Access
- Reconciliation

Unlike disconnected applications, administrator intervention is not required for routine operations.

Simple Definition

A **Connected Application** is an application that communicates directly with Saviynt through a connector, allowing automated identity and access management.

Why are Connected Applications Used?

Organizations use connected applications because automation reduces manual work and improves security.

Instead of administrators manually creating and deleting accounts, Saviynt performs these tasks automatically.

Benefits

Automated Provisioning

Accounts are created automatically after approvals.

Automated Deprovisioning

Accounts are disabled automatically when users leave.

Automatic Reconciliation

Saviynt imports the latest accounts and entitlements from the application.

Reduced Manual Effort

IT teams spend less time managing accounts.

Better Security

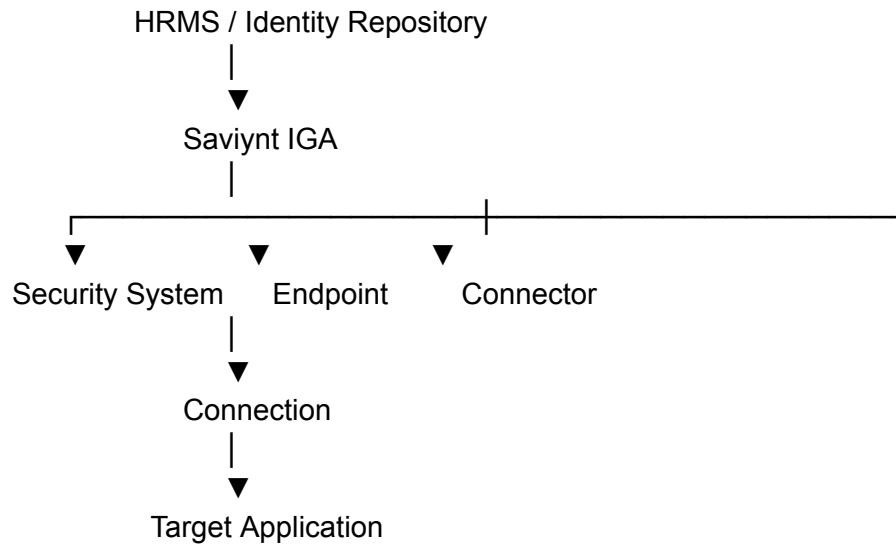
Access is granted and removed consistently.

Compliance

Supports audit requirements by maintaining complete logs of identity and access changes.

Connected Application Architecture

A connected application consists of several components working together.



Components

- Security System
- Endpoint
- Connector
- Connection
- Target Application

Each component has a specific responsibility.

What is a Security System?

Definition

A **Security System** is the logical representation of an application inside Saviynt.

It defines **what application is being governed**.

It acts as the parent object that groups related endpoints, accounts, entitlements, and governance policies.

Simple Definition

Security System is the identity of an application inside Saviynt.

Why is a Security System Needed?

Without a Security System, Saviynt would not know:

- Which application is being managed.
- Which endpoint belongs to which application.
- Which accounts belong to that application.
- Which entitlements belong to it.

It provides organization and governance.

Responsibilities of a Security System

- Represents an application.
 - Stores application information.
 - Links endpoints.
 - Links connectors.
 - Stores application accounts.
 - Stores entitlements.
 - Enables governance.
-

One-to-One Security System

Definition

One Security System is connected to one Endpoint.

Example:

Security System

↓

Active Directory

↓

One Endpoint

When is it Used?

Used when:

- Only one environment exists.
 - One production server.
 - Simple architecture.
-

Example

One Active Directory domain

↓

One Endpoint

↓

One Security System

One-to-Many Security System

Definition

One Security System can contain multiple Endpoints.

Example:

Security System

↓

SAP

├— Production

├— Test

└— Development

Why?

Organizations often have multiple environments.

Instead of creating three Security Systems,

Saviynt creates:

One Security System

Multiple Endpoints.

Advantages

- Easier management.
 - Shared governance.
 - Common reporting.
 - Better organization.
-

Real Example

Company uses AWS.

AWS has:

- Development Account
- Test Account
- Production Account

Saviynt creates:

Security System

↓

AWS

↓

Three Endpoints

Development

Test

Production

All belong to the same Security System.

Part 2 – Endpoint and Connector

What is an Endpoint?

Definition

An **Endpoint** represents a specific connection to an application.

It contains all connection details required to communicate with the application.

Simple Definition

Endpoint is the actual connection between Saviynt and the target application.

What Information Does an Endpoint Store?

- Target URL
 - Connector Type
 - Authentication Method
 - Username
 - Password
 - SSL Configuration
 - API Details
 - Connection Parameters
-

Why is an Endpoint Needed?

The Security System only represents the application.

The Endpoint performs the communication.

Without an Endpoint:

Saviynt cannot:

- Connect.
 - Import accounts.
 - Provision users.
 - Perform reconciliation.
-

Endpoint Workflow

Security System



Endpoint



Connector



Target Application

Endpoint vs Security System

Security System	Endpoint
Represents the application	Represents the connection
Logical object	Technical object
Stores application details	Stores connection details
Used for governance	Used for communication
One Security System can have many Endpoints	One Endpoint belongs to one Security System

Multiple Endpoints

Organizations often maintain:

- Development
- Testing
- UAT
- Production

Each environment requires its own Endpoint.

Example:

Security System

↓

Salesforce

└─ Dev Endpoint

└─ Test Endpoint

└─ UAT Endpoint

└─ Production Endpoint

Benefits

- Separate configurations.
 - Safer testing.
 - Independent credentials.
 - Independent URLs.
-

Real Example

Company has three SAP systems.

- SAP DEV
- SAP QA
- SAP PROD

Saviynt creates:

One Security System

↓

SAP



Three Endpoints

Each endpoint connects to its respective SAP environment.

What is a Connector?

Definition

A **Connector** is software that enables communication between Saviynt and the target application.

It understands how to:

- Authenticate.
 - Read data.
 - Import accounts.
 - Provision users.
 - Remove access.
 - Perform reconciliation.
-

Simple Definition

Connector is the communication bridge between Saviynt and an application.

Why is a Connector Needed?

Every application communicates differently.

Examples:

- REST API

- SOAP API
- LDAP
- Database

The connector knows how to communicate using the application's supported protocol.

Connector Workflow

Saviynt



Connector



Application API



Application

OOTB (Out-of-the-Box) Connector

Definition

OOTB Connectors are pre-built connectors provided by Saviynt.

Examples:

- Active Directory
- Azure AD
- AWS
- Salesforce
- SAP
- Oracle
- Workday

Advantages

- Ready to use.
 - Minimal configuration.
 - Fully tested.
 - Vendor supported.
 - Faster implementation.
-

Custom Connector

Definition

A Custom Connector is developed when no OOTB connector exists.

Organizations create custom integrations using:

- REST APIs
 - SOAP APIs
 - Java
 - SCF
-

When is it Used?

- Homegrown applications.
 - Legacy systems.
 - Proprietary applications.
-

Advantages

- Highly flexible.
 - Supports custom business requirements.
-

Disadvantages

- Higher development effort.
 - More maintenance.
 - Longer implementation time.
-

SCF (Saviynt Connector Framework)

Definition

SCF (Saviynt Connector Framework) is a framework that helps developers build custom connectors for applications that do not have OOTB support.

It provides reusable components for:

- Authentication
 - API communication
 - Import jobs
 - Provisioning
 - Reconciliation
-

Benefits

- Standard development approach.
 - Easier maintenance.
 - Reusable connector logic.
 - Faster custom connector development.
-

REST Connector

Definition

A REST Connector communicates with applications that expose REST APIs.

It exchanges data using:

- HTTP/HTTPS
 - JSON
-

Example

Saviynt sends a REST API request to create a Salesforce user.

SOAP Connector

Definition

A SOAP Connector communicates with applications that expose SOAP web services.

It exchanges data using:

- XML
 - SOAP Messages
-

Example

Legacy enterprise applications often use SOAP services for provisioning.

LDAP Connector

Definition

An LDAP Connector communicates with directory services that support the LDAP protocol.

Common targets include:

- Active Directory
 - OpenLDAP
-

Typical Operations

- Import users.
 - Import groups.
 - Provision accounts.
 - Disable users.
-

Database Connector

Definition

A Database Connector communicates directly with relational databases.

Supported databases may include:

- Oracle
- SQL Server
- MySQL
- PostgreSQL

It retrieves or updates identity data using SQL queries.

Use Cases

- Import users from a database.
 - Import entitlements.
 - Synchronize account information.
 - Support legacy applications without APIs.
-

Real Project Example

ABC Bank wants to onboard Salesforce.

1. Administrator creates a **Security System** named "Salesforce".
 2. A **Production Endpoint** is created with the Salesforce URL and credentials.
 3. The **Salesforce OOTB Connector** is selected.
 4. The connection is tested successfully.
 5. Saviynt imports accounts and entitlements.
 6. Provisioning and reconciliation are enabled.
 7. Users can now request Salesforce access through Saviynt.
-

Best Practices

- Use OOTB connectors whenever available.
 - Use one Security System per application.
 - Create separate Endpoints for Development, Test, UAT, and Production.
 - Store credentials securely.
 - Test every Endpoint before enabling provisioning.
 - Use meaningful names for Security Systems and Endpoints.
 - Monitor connector logs for authentication or communication issues.
 - Document connector configurations for future maintenance.
-

Interview Questions

Q1. What is a Connected Application?

A Connected Application is an application that integrates directly with Saviynt through a connector, enabling automated provisioning, reconciliation, and identity governance.

Q2. What is the difference between a Security System and an Endpoint?

A Security System is the logical representation of an application in Saviynt, while an Endpoint contains the technical connection details required to communicate with that application.

Q3. Why can one Security System have multiple Endpoints?

Because organizations often have separate Development, Test, UAT, and Production environments, each requiring its own connection while still representing the same application.

Q4. What is a Connector?

A Connector is the communication component that allows Saviynt to interact with a target application for importing data, provisioning accounts, and performing reconciliation.

Q5. Why are OOTB Connectors preferred?

They are pre-built, tested, vendor-supported, require less configuration, and reduce implementation time compared to custom connectors.

Q6. What is SCF?

SCF (Saviynt Connector Framework) is a framework used to build custom connectors when no Out-of-the-Box connector is available.

Q7. When would you use a Database Connector?

When user or account information is stored in a relational database and no API or standard connector is available.

Quick Revision

- A **Connected Application** integrates directly with Saviynt.
- A **Security System** logically represents the application.
- An **Endpoint** stores the technical connection details.
- One Security System can have multiple Endpoints for different environments.
- A **Connector** enables communication with the target application.

- **OOTB Connectors** are preferred over custom connectors.
- **SCF** is used to develop custom connectors.
- Saviynt supports REST, SOAP, LDAP, and Database connectors based on the application's integration method.

Module 3 – Connected Application Onboarding (Part 3 & Part 4)

Part 3 – Connection and Application Enrollment

What is a Connection?

Definition

A **Connection** is the configuration that allows Saviynt to communicate with a target application through a connector.

It contains all the technical information required to establish communication between Saviynt and the application.

Without a valid connection, Saviynt cannot:

- Import accounts
- Import entitlements
- Provision users
- Disable users
- Perform reconciliation

Simple Definition

A **Connection** is the communication setup between Saviynt and a target application.

Why is a Connection Required?

Saviynt must know:

- Where the application is located.
- How to connect.
- How to authenticate.
- Which connector to use.

Without this information, no communication is possible.

Components of a Connection

A connection usually contains:

- Target URL
 - Connector Type
 - Authentication Method
 - Username
 - Password
 - API Token (if required)
 - SSL Certificate Information
 - Timeout Settings
-

Connection Workflow

Administrator Creates Connection

|



Configure Target URL

|



Configure Authentication

|



Provide Credentials

|



Test Connection



Save Connection



Saviynt Communicates with Application

Authentication

Definition

Authentication is the process of verifying Saviynt's identity before allowing it to communicate with the target application.

The target application checks whether Saviynt is authorized.

Common Authentication Methods

Username and Password

Most common authentication method.

Example:

Administrator account

Service account

API Token

Used by cloud applications.

Instead of passwords,

applications use secure API Tokens.

OAuth

Commonly used for cloud applications.

Examples:

- Salesforce
 - Microsoft 365
 - Google Workspace
-

Certificate-Based Authentication

Uses SSL certificates instead of passwords.

Provides stronger security.

Why Authentication is Important?

Without authentication:

- Anyone could access the application.
- Sensitive data could be exposed.
- Unauthorized provisioning could occur.

Authentication protects communication.

Target URL

Definition

The **Target URL** is the address of the application that Saviynt connects to.

It tells Saviynt where the application is located.

Example

For Salesforce:

`https://company.my.salesforce.com`

For Workday:

`https://company.workday.com`

Why is the Target URL Required?

Saviynt must know:

- Which server to contact.
- Which environment to use.
- Which API endpoint to access.

Without the correct URL, the connection fails.

Credentials

Definition

Credentials are the authentication details used by Saviynt to access the application.

Common Credentials

- Username
 - Password
 - API Key
 - Client ID
 - Client Secret
 - OAuth Token
 - SSL Certificate
-

Best Practices

- Use dedicated service accounts.
 - Do not use personal user accounts.
 - Rotate passwords regularly.
 - Store credentials securely.
 - Grant minimum required permissions.
-

Connection Testing

Definition

Connection Testing verifies that Saviynt can successfully communicate with the target application.

What Happens During Testing?

Saviynt checks:

- Network connectivity.
- Authentication.
- API availability.
- Credentials.

- SSL certificates.
 - Connector configuration.
-

Why Test the Connection?

Testing identifies configuration issues before onboarding begins.

It helps avoid failures during:

- User Import
 - Provisioning
 - Reconciliation
-

Common Test Failures

- Wrong URL
 - Invalid credentials
 - Expired password
 - SSL certificate issue
 - Firewall blocking
 - API unavailable
 - Network timeout
-

What is Application Enrollment?

Definition

Application Enrollment is the process of registering a new application in Saviynt and configuring it for Identity Governance.

Enrollment creates the basic application structure required for onboarding.

Why is Enrollment Required?

Enrollment prepares Saviynt to manage the application.

It creates the initial configuration before provisioning and reconciliation are enabled.

Rapid Enrollment

Definition

Rapid Enrollment is a quick onboarding method that uses predefined templates and default settings.

When is it Used?

- Standard applications.
 - OOTB connectors.
 - Simple configurations.
 - Fast deployments.
-

Advantages

- Faster implementation.
 - Less manual configuration.
 - Lower chance of configuration errors.
-

Assisted Configuration

Definition

Assisted Configuration guides administrators through each configuration step using a wizard.

Advantages

- Easy for beginners.
 - Step-by-step guidance.
 - Reduces configuration mistakes.
 - Recommended for new implementations.
-

Advanced Configuration

Definition

Advanced Configuration allows administrators to customize every aspect of the application.

When is it Used?

- Complex environments.
 - Custom provisioning.
 - Advanced workflows.
 - Organization-specific requirements.
-

Advantages

- Maximum flexibility.
 - Supports custom business logic.
 - Fine-grained control.
-

Disadvantages

- More configuration effort.
 - Requires experienced administrators.
-

Feature Selection

Definition

During enrollment, administrators choose which Saviynt features should be enabled for the application.

Common Features

- Account Import
 - Entitlement Import
 - User Import
 - Provisioning
 - Deprovisioning
 - Password Management
 - Reconciliation
 - Certifications
 - Access Requests
-

Why Feature Selection is Important?

Not every application requires every feature.

Enabling only the required features:

- Improves performance.
 - Reduces complexity.
 - Simplifies maintenance.
-

Part 4 – New Application Onboarding

What is New Application Onboarding?

New Application Onboarding is the process of adding a completely new application into Saviynt for Identity Governance.

Once onboarded,

Saviynt can manage:

- Accounts
 - Entitlements
 - Access Requests
 - Provisioning
 - Reconciliation
-

Application Catalog

Definition

The Application Catalog is the list of applications available for onboarding.

It includes:

- Supported OOTB applications.
 - Previously onboarded applications.
 - Marketplace applications.
-

Benefits

- Easy application selection.
 - Standardized onboarding.
 - Faster implementation.
-

Saviynt Exchange Marketplace

Definition

Saviynt Exchange Marketplace is a repository of pre-built integrations, connectors, templates, and application configurations.

Administrators can browse and use these resources instead of creating integrations from scratch.

Advantages

- Ready-made connectors.
 - Faster deployment.
 - Vendor-supported integrations.
 - Reduced implementation effort.
-

Create Application

Definition

If the required application is not already available, administrators can manually create a new application.

Typical information includes:

- Application Name
 - Description
 - Owner
 - Connector Type
 - Authentication Method
 - Environment
-

Browse Marketplace

Administrators can search the Saviynt Exchange Marketplace by:

- Application Name
- Vendor
- Connector Type
- Category

Example:

Search:

Salesforce

Results:

- Salesforce Connector
 - Provisioning Templates
 - Import Jobs
-

Configuration

After selecting or creating the application,

administrators configure:

- Connection
- Endpoint
- Authentication
- Connector
- Attribute Mapping
- Import Jobs
- Provisioning Settings
- Reconciliation Settings

Only after configuration is complete can the application be governed.

Post Enrollment Check

Definition

After enrollment, administrators verify that the application has been configured correctly.

Typical Checks

- Connection created.
 - Endpoint created.
 - Security System created.
 - Connector assigned.
 - Authentication successful.
 - Connection test successful.
 - Import jobs available.
 - Provisioning configuration complete.
-

Why Does Saviynt Automatically Create These Objects?

When a connected application is enrolled, Saviynt automatically creates several objects because they are required for governance.

Connection

Created automatically because Saviynt must know how to communicate with the application.

It stores:

- URL
 - Authentication
 - Credentials
-

Endpoint

Created automatically because every application requires a technical connection.

The endpoint contains:

- Connector
 - Connection parameters
 - API information
-

Security System

Created automatically because Saviynt must have a logical representation of the application.

The Security System groups:

- Endpoints
 - Accounts
 - Entitlements
 - Governance configuration
-

Why Are These Objects Linked?

These objects work together as one complete configuration.

Security System

|



Endpoint

|



Connection

|



Connector

|



Target Application

Relationship

- **Security System** identifies the application.
- **Endpoint** defines the technical environment.
- **Connection** stores communication details.
- **Connector** knows how to interact with the application.

Without any one of these objects, Saviynt cannot successfully manage the application.

Real Project Example

ABC Bank wants to onboard Microsoft 365.

Step 1

The administrator searches the **Saviynt Exchange Marketplace** and selects the Microsoft 365 OOTB connector.

Step 2

Rapid Enrollment is chosen.

Step 3

The administrator enters:

- Target URL
- OAuth credentials
- Tenant information

Step 4

The connection is tested successfully.

Step 5

Saviynt automatically creates:

- Security System
- Production Endpoint
- Connection

Step 6

The administrator enables:

- Account Import
- Entitlement Import
- Provisioning
- Reconciliation

Step 7

The application is ready for identity governance.

Best Practices

- Use Rapid Enrollment for standard OOTB applications.
 - Use Assisted Configuration if you are new to the application.
 - Use Advanced Configuration only when customization is required.
 - Always test the connection before saving.
 - Use service accounts instead of personal accounts.
 - Verify the Target URL and credentials.
 - Enable only the features required for the application.
 - Perform a Post Enrollment Check before enabling production jobs.
 - Document connection details and authentication methods.
-

Interview Questions

Q1. What is a Connection?

A Connection is the configuration that allows Saviynt to communicate securely with a target application using a connector.

Q2. Why is Authentication required?

Authentication verifies Saviynt's identity before the target application allows communication, protecting against unauthorized access.

Q3. What is the purpose of the Target URL?

The Target URL tells Saviynt where the target application is located and which API endpoint to use.

Q4. What is Application Enrollment?

Application Enrollment is the process of registering and configuring a new application in Saviynt so that it can be managed through identity governance.

Q5. What is the difference between Rapid Enrollment and Advanced Configuration?

Rapid Enrollment uses predefined templates for quick setup, while Advanced Configuration provides full customization for complex environments.

Q6. What is a Post Enrollment Check?

A Post Enrollment Check verifies that the Connection, Endpoint, Security System, Connector, and required configurations have been created successfully before the application is used in production.

Q7. Why does Saviynt automatically create the Connection, Endpoint, and Security System?

Because all three are essential for application governance. The Security System represents the application, the Endpoint defines the technical environment, and the Connection stores the communication details required by the Connector.

Quick Revision

- A **Connection** stores the communication settings between Saviynt and the target application.
- **Authentication** verifies Saviynt before communication begins.
- The **Target URL** identifies the application's location.
- **Credentials** allow secure access to the application.
- **Connection Testing** validates connectivity and configuration.
- **Application Enrollment** registers an application for governance.
- Enrollment options include **Rapid**, **Assisted**, and **Advanced** configuration.
- The **Application Catalog** and **Saviynt Exchange Marketplace** simplify onboarding.
- After enrollment, Saviynt automatically creates the **Security System**, **Endpoint**, and **Connection** because all are required for successful integration and governance.

Module 3 – Part 5A

Saviynt Data Model Fundamentals

What is the Saviynt Data Model?

Definition

The **Saviynt Data Model** defines how all identity objects are organized and related inside Saviynt.

It shows the relationship between:

- Connector
- Security System
- Endpoint
- Users
- Accounts
- Entitlement Types
- Entitlement Values

The Data Model helps Saviynt understand:

- Who the users are.
 - Which accounts belong to them.
 - Which applications they use.
 - What access they have.
 - How provisioning and reconciliation should work.
-

Simple Definition

The Saviynt Data Model is the internal structure that defines how identities, applications, accounts, and access are connected inside Saviynt.

Why is the Saviynt Data Model Important?

Every Saviynt feature depends on this model.

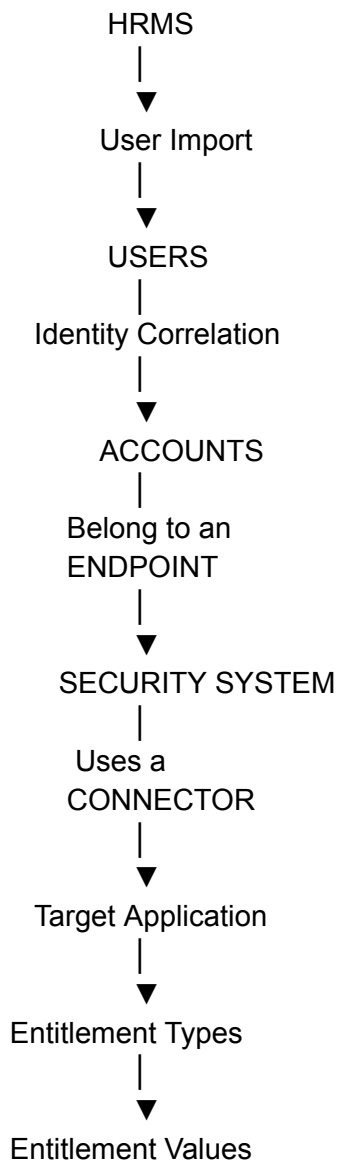
Without it, Saviynt cannot:

- Import users.
 - Import accounts.
 - Correlate identities.
 - Provision accounts.
 - Assign entitlements.
 - Perform certifications.
 - Generate audit reports.
-

Why Interviewers Ask About It

Understanding the Data Model shows that you understand **how Saviynt works internally**, not just how to click buttons in the UI.

Complete Saviynt Data Model



This diagram represents the complete relationship between all major Saviynt objects.

Connector

Definition

A **Connector** is the communication component that connects Saviynt with the target application.

It understands how to communicate with the application using supported protocols.

Examples:

- REST
 - SOAP
 - LDAP
 - Database
-

Responsibilities

A Connector can:

- Authenticate
 - Import Accounts
 - Import Entitlements
 - Provision Users
 - Disable Users
 - Update Accounts
 - Perform Reconciliation
-

Example

Salesforce Connector

↓

Connects Saviynt with Salesforce

↓

Imports Users

↓

Creates Accounts

↓

Assigns Roles

Security System

Definition

A **Security System** is the logical representation of an application inside Saviynt.

Every onboarded application has one Security System.

Responsibilities

The Security System:

- Represents the application.
 - Groups Endpoints.
 - Stores Accounts.
 - Stores Entitlements.
 - Applies Governance Policies.
-

Example

Application

Salesforce

↓

Security System

Salesforce

Everything related to Salesforce belongs to this Security System.

Endpoint

Definition

An **Endpoint** is the technical connection between Saviynt and the application.

It stores:

- URL
 - Credentials
 - Authentication
 - Connector Information
-

Why is Endpoint Needed?

The Security System represents the application.

The Endpoint connects to it.

Without an Endpoint,

Saviynt cannot communicate with the application.

Example

Security System

Salesforce

↓

Production Endpoint

↓

Users

Definition

Users represent identities inside Saviynt.

Users are imported from:

- Workday
 - SAP SuccessFactors
 - Oracle HCM
 - Active Directory
 - CSV
 - REST API
-

Information Stored

Typical user attributes include:

- Employee ID
 - Username
 - First Name
 - Last Name
 - Email
 - Department
 - Manager
 - Status
-

Example

Employee ID	Username
1001	rahul

Accounts

Definition

An **Account** is a user account that exists in a target application.

One User can have multiple Accounts.

Example

User

Rahul

may have:

- Active Directory Account
- SAP Account
- AWS Account
- Salesforce Account

One identity,

Multiple accounts.

Relationship

User

↓

Multiple Accounts

↓

User vs Account

User	Account
Identity inside Saviynt	Login inside an application
Imported from HR	Imported from Applications
One identity	Many accounts
Central identity	Application-specific

Example

User

Rahul Sharma

Accounts

rahul.ad

rahul.sap

rahul.aws

rahul.salesforce

All belong to the same identity.

Entitlement Types

Definition

An **Entitlement Type** defines the category of access available in an application.

It answers the question:

"What kind of access can be assigned?"

Examples

Active Directory

Entitlement Type

Group

AWS

Entitlement Type

IAM Role

SAP

Entitlement Type

Role

Salesforce

Entitlement Type

Profile

Why is Entitlement Type Important?

It organizes similar permissions into categories.

Entitlement Values

Definition

An **Entitlement Value** is the actual permission assigned to a user.

It answers the question:

"Which specific access is granted?"

Examples

Entitlement Type

Group

↓

Entitlement Values

Finance_Group

HR_Group

IT_Group

AWS

Entitlement Type

IAM Role

↓

Entitlement Values

Administrator

Developer

ReadOnly

Salesforce

Entitlement Type

Profile

↓

Entitlement Values

System Administrator

Sales User

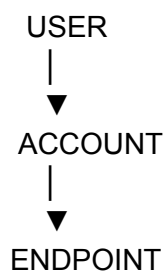
Marketing User

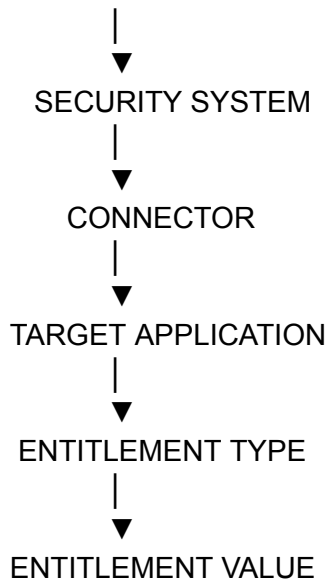
Entitlement Type vs Entitlement Value

Entitlement Type	Entitlement Value
Category	Actual Permission
Group	Finance_Group
Role	SAP Finance
IAM Role	Administrator

Relationship Between All Objects

This is one of the most important interview topics.





Explanation of Relationships

User → Account

One User can have many Accounts.

Example:

Rahul

↓

AD

SAP

AWS

Account → Endpoint

Every account belongs to an Endpoint.

Example

SAP Account



SAP Production Endpoint

Endpoint → Security System

Every Endpoint belongs to one Security System.

Example

SAP DEV

SAP QA

SAP PROD



SAP Security System

Security System → Connector

Every Security System uses one Connector.

Connector enables communication.

Application → Entitlement Type

Each application defines its own categories of access.

Example

Active Directory



Group

Entitlement Type → Entitlement Value

Example

Group



Finance

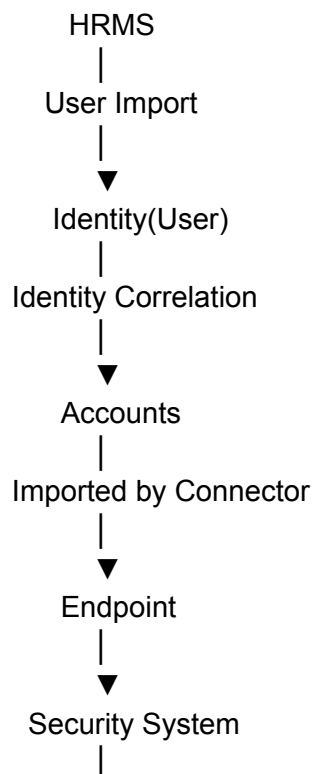


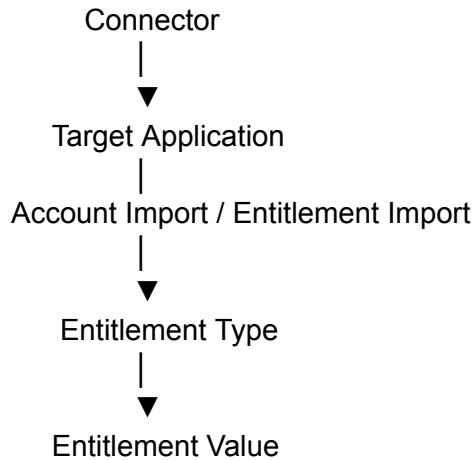
HR



IT

Complete Internal Architecture





Real Example – Active Directory

Application

Active Directory



Security System

Active Directory



Endpoint

AD Production



Connector

LDAP Connector



Entitlement Type

Group

↓

Entitlement Values

Finance_Group

HR_Group

IT_Group

↓

User Rahul

↓

AD Account

↓

Assigned

Finance_Group

Real Example – AWS

Application

AWS

↓

Security System

AWS

↓

Endpoint

Production AWS

↓

Connector

AWS Connector



Entitlement Type

IAM Role



Entitlement Values

Administrator

Developer

ReadOnly



User Priya



AWS Account



Assigned

Developer Role

Key Points to Remember

- **User** = Identity inside Saviynt.
- **Account** = User account in a target application.
- **Security System** = Logical representation of the application.
- **Endpoint** = Technical connection to the application.
- **Connector** = Communication bridge between Saviynt and the application.
- **Entitlement Type** = Category of access.
- **Entitlement Value** = Specific permission assigned to the user.
- One **User** can have many **Accounts**.

- One **Security System** can have many **Endpoints**.
 - One **Entitlement Type** can contain many **Entitlement Values**.
-

Best Practices

- Use one Security System per application.
 - Create separate Endpoints for Development, Test, UAT, and Production.
 - Prefer Out-of-the-Box (OOTB) connectors whenever possible.
 - Maintain consistent naming conventions for Security Systems, Endpoints, and Connectors.
 - Keep Entitlement Types broad (for example, Group or Role) and Entitlement Values specific (for example, Finance_Group or SAP_FI_Admin).
 - Ensure User-to-Account correlation is accurate to prevent duplicate identities.
 - Regularly reconcile accounts and entitlements to keep Saviynt synchronized with target applications.
-

Quick Revision

- The **Saviynt Data Model** defines the relationships between all identity and application objects.
- **Connector** communicates with the application.
- **Security System** represents the application inside Saviynt.
- **Endpoint** stores the technical connection details.
- **Users** are identities imported into Saviynt.
- **Accounts** are application-specific login accounts linked to users.
- **Entitlement Types** categorize access.
- **Entitlement Values** are the actual permissions granted.
- Understanding the relationships between these objects is essential for application onboarding, provisioning, reconciliation, and identity governance.

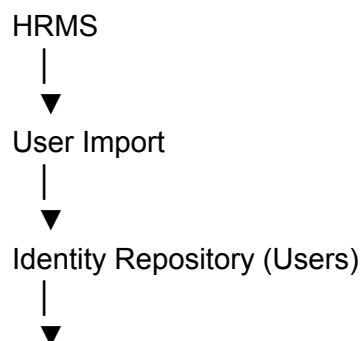
Module 3 – Part 5B

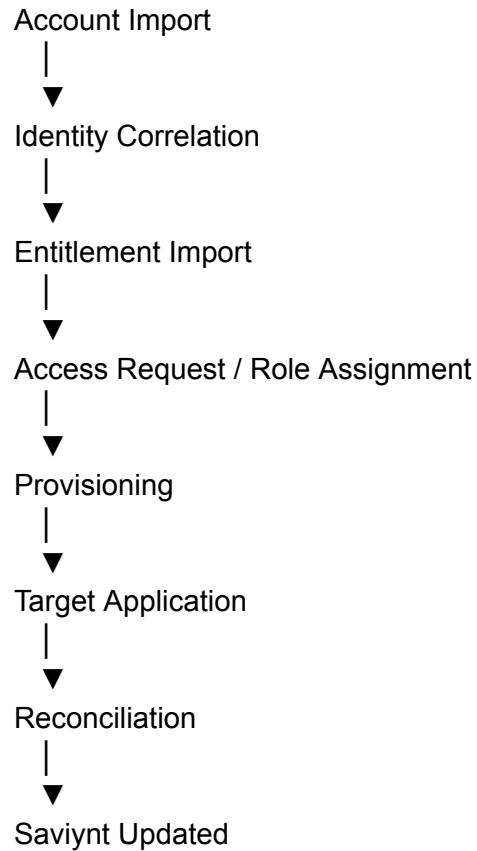
Internal Workflow & Real Project Examples

Complete Internal Workflow

Once an application is onboarded, Saviynt follows a sequence of steps to manage identities and application access.

The complete workflow is:





Each step is explained below.

Step 1 – User Import

Definition

The User Import Job imports employee identities from the Authoritative Source into Saviynt.

The source may be:

- Workday
- SAP SuccessFactors
- Oracle HCM
- Active Directory
- CSV
- REST API

Information Imported

- Employee ID
- Username
- Email
- Department
- Manager
- Status
- Job Title

What Happens?

Saviynt validates the data and creates identities in the **Identity Repository**.

Example

Workday contains:

Rahul Sharma

↓

Saviynt imports Rahul.

↓

Identity Created.

Step 2 – Identity Repository

After User Import,

the identity is stored inside Saviynt.

Example

Employee ID : 1001
Username : rahul
Department : Finance
Status : Active

At this stage,

Rahul exists only as an identity.

No application accounts have been linked yet.

Step 3 – Account Import

Definition

Account Import retrieves existing accounts from target applications.

Direction:

Application

↓

Saviynt

Why?

Some users may already have accounts before Saviynt was implemented.

Saviynt imports those accounts so it can govern them.

Example

Active Directory contains

Username

rahul

Saviynt imports:

AD Account

↓

Links it later to Rahul.

Step 4 – Identity Correlation

Definition

Correlation links imported application accounts to the correct Saviynt identity.

This is one of the most important processes in Identity Governance.

How Correlation Works

Saviynt compares attributes such as:

- Employee ID
- Username
- Email
- Employee Number

If they match,

the account is linked to the identity.

Example

Identity

rahul

Imported Account

rahul

Result

Rahul Identity

↓

Rahul AD Account

Successfully correlated.

What Happens if Correlation Fails?

Example

Identity

rahul

Account

rahul.sharma

Saviynt cannot correlate.

The account becomes an **orphan account** until correlation rules are corrected.

Step 5 – Entitlement Import

Definition

Entitlement Import imports permissions available in the application.

These permissions become requestable inside Saviynt.

Examples

Active Directory

Entitlement Type

Group

↓

Entitlement Values

- Finance_Group
 - HR_Group
 - IT_Group
-

AWS

Entitlement Type

IAM Role

↓

Entitlement Values

- Administrator
 - Developer
 - ReadOnly
-

SAP

Entitlement Type

Role



Entitlement Values

- SAP_FI_User
 - SAP_HR_User
-

Why Import Entitlements?

Without entitlements,

users cannot request application access.

Step 6 – Access Request / Role Assignment

After users, accounts, and entitlements exist,

Saviynt can grant access.

Access may be granted through:

- Business Roles
 - Access Requests
 - Birthright Rules
 - Dynamic Rules
-

Example

Rahul belongs to Finance.

Finance Business Role includes:

- AD Finance Group
- SAP Finance Role
- Microsoft 365 License

Rahul automatically receives those entitlements.

Step 7 – Provisioning

Definition

Provisioning creates or updates accounts in the target application.

Direction:

Saviynt



Target Application

Provisioning Operations

Saviynt can:

- Create account
 - Enable account
 - Disable account
 - Update attributes
 - Assign group
 - Remove group
 - Reset password
 - Assign role
-

Example

Rahul requests AWS access.

Manager approves.

Saviynt provisions:

AWS Account



Developer IAM Role

Automatically.

Step 8 – Reconciliation

Definition

Reconciliation imports the latest account and entitlement information from the target application back into Saviynt.

Direction:

Application



Saviynt

Purpose

Verify that the requested changes actually occurred.

Detect changes made outside Saviynt.

Keep Saviynt synchronized with the application.

Example

Administrator directly adds Rahul to:

Finance_Group

in Active Directory.

Reconciliation imports the latest AD information.

Saviynt updates Rahul's access record.

Complete End-to-End Workflow

HR Creates Employee



Workday



User Import



Identity Repository



Account Import



Identity Correlation



Entitlement Import



Access Request



Approval

↓

Provisioning

↓

Application Updated

↓

Reconciliation

↓

Saviynt Updated

AWS Example

Company uses AWS.

Step 1

Application onboarded.

AWS Connector configured.

Step 2

AWS Account Import runs.

Imported Accounts

↓

rahul.aws

↓

priya.aws

Step 3

IAM Roles imported.

Entitlement Type

IAM Role

Entitlement Values

- Administrator
- Developer
- ReadOnly

Step 4

Correlation

Rahul Identity

↓

rahul.aws Account

Step 5

Rahul requests Developer Role.

Approval received.

Provisioning assigns:

Developer IAM Role.

Step 6

Nightly Reconciliation verifies the role assignment.

Saviynt confirms AWS matches the requested access.

Active Directory Example

Application

Active Directory

Step 1

LDAP Connector imports AD accounts.

Step 2

Group Import runs.

Entitlement Type

Group

Entitlement Values

- Finance_Group
 - HR_Group
 - VPN_Group
-

Step 3

Correlation

Rahul Identity

↓

rahul AD Account

Step 4

Rahul requests VPN access.

Approval completed.

Provisioning adds Rahul to:

VPN_Group

Step 5

Next reconciliation verifies Rahul is now a member of VPN_Group.

Real Project Example

ABC Bank implements Saviynt.

Applications:

- Workday
 - Active Directory
 - SAP
 - Salesforce
 - AWS
-

Joiner

HR hires Priya.

Workday is updated.

User Import Job creates Priya's identity.

Account Import identifies existing Microsoft 365 and AD accounts.

Correlation links those accounts to Priya.

Entitlement Import retrieves available groups and roles.

Finance Business Role is assigned.

Provisioning creates:

- SAP account
- Salesforce account
- AWS account

Priya receives all required access before her first working day.

Mover

Priya transfers from Finance to HR.

Workday updates the department.

User Import updates the identity.

Business Role changes.

Provisioning removes Finance access and grants HR access.

Reconciliation confirms the changes.

Leaver

Priya resigns.

Workday changes the status to Terminated.

User Import updates the identity.

Provisioning disables:

- AD account
- SAP account

- Salesforce account
- AWS account

Reconciliation verifies all accounts are disabled.

Audit logs record every action.

End-to-End Onboarding Scenario

A new application, Salesforce, is introduced.

Step 1 – Discover

Salesforce is identified during Application Discovery.

Step 2 – Onboard

Administrator selects the Salesforce OOTB Connector.

Saviynt creates:

- Security System
 - Endpoint
 - Connection
-

Step 3 – Configure

Connection details are entered.

Connection Test succeeds.

Step 4 – Import

Account Import retrieves Salesforce accounts.

Entitlement Import retrieves:

- Profiles
 - Roles
 - Permission Sets
-

Step 5 – Correlation

Imported Salesforce accounts are linked to existing identities using username or employee ID.

Step 6 – Governance

Users can now:

- Request Salesforce access.
 - Receive approvals.
 - Be provisioned automatically.
 - Participate in certifications.
 - Have their access audited.
-

Step 7 – Continuous Operations

Scheduled jobs continue to run:

- User Import
- Account Import
- Entitlement Import
- Reconciliation

This ensures Saviynt and Salesforce remain synchronized.

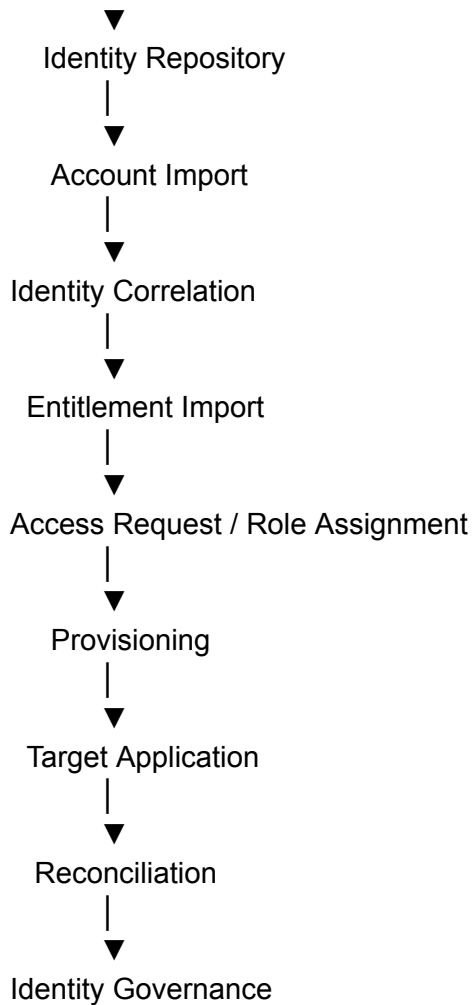
Internal Workflow Summary

Authoritative Source (HRMS)



User Import





Key Takeaways

- **User Import** creates identities in Saviynt.
- **Account Import** imports application accounts.
- **Correlation** links accounts to identities.
- **Entitlement Import** imports requestable permissions.
- **Provisioning** pushes approved access to target applications.
- **Reconciliation** validates and synchronizes application data back into Saviynt.
- Together, these processes provide complete identity governance from onboarding to deprovisioning.

Module 3 – Part 5C

Best Practices, Troubleshooting & Interview Questions

Best Practices

Following best practices helps ensure successful application onboarding, improves performance, simplifies maintenance, and reduces production issues.

1. Use Out-of-the-Box (OOTB) Connectors Whenever Possible

OOTB connectors are developed and tested by Saviynt.

They provide:

- Automatic provisioning
- Automatic reconciliation
- Vendor support
- Easier upgrades
- Lower maintenance

Use custom connectors only when no OOTB connector is available.

2. Use One Security System Per Application

Each application should normally have one Security System.

Example:

- Salesforce → Salesforce Security System
- AWS → AWS Security System

Avoid creating multiple Security Systems for the same application unless there is a valid business requirement.

3. Create Separate Endpoints for Different Environments

Organizations usually maintain:

- Development
- Test
- UAT
- Production

Each environment should have its own Endpoint.

Example:

AWS Security System

|— AWS DEV Endpoint

|— AWS TEST Endpoint

|— AWS UAT Endpoint

└— AWS PROD Endpoint

Benefits:

- Safer testing
 - Independent credentials
 - Easier troubleshooting
 - Better change management
-

4. Use Dedicated Service Accounts

Always configure connections using dedicated service accounts.

Do not use personal administrator accounts.

Benefits:

- Better security
 - Easier auditing
 - Password changes do not affect personal users
 - Easier credential management
-

5. Test Every Connection Before Production

Always perform a successful connection test before enabling:

- Account Import
- Entitlement Import
- Provisioning
- Reconciliation

This helps identify:

- Incorrect URLs
- Authentication failures
- SSL issues
- Network problems

6. Use Meaningful Naming Conventions

Use consistent names.

Examples:

Security System

Salesforce

AWS

ActiveDirectory

Endpoint

Salesforce_PROD

AWS_DEV

SAP_QA

Connector

Salesforce_OOTB

LDAP_Connector

Consistent naming simplifies administration.

7. Keep Entitlements Organized

Maintain:

- Clear Entitlement Types
- Meaningful Entitlement Values

Example

Entitlement Type

Group

Entitlement Values

Finance_Group

HR_Group

IT_Group

8. Schedule Regular Reconciliation

Reconciliation keeps Saviynt synchronized with applications.

Without reconciliation:

- Access becomes outdated.
 - Reports become inaccurate.
 - Certifications may fail.
-

9. Validate Correlation Rules

Verify that imported accounts correctly match identities.

Incorrect correlation creates orphan accounts and duplicate identities.

10. Test in Lower Environments First

Always test configuration changes in:

- Development
- Test
- UAT

before Production.

Common Configuration Mistakes

1. Wrong Target URL

Example

Production URL entered instead of Test URL.

Result

Connection fails.

2. Incorrect Credentials

Example

Expired password.

Disabled service account.

Authentication fails.

3. Incorrect Endpoint Mapping

Endpoint configured for the wrong application.

Provisioning may update incorrect systems.

4. Duplicate Security Systems

Creating multiple Security Systems for one application leads to:

- Duplicate accounts
 - Reporting confusion
 - Governance issues
-

5. Incorrect Correlation Rules

Example

User

rahul

Application Account

rahul.sharma

Correlation fails.

Orphan account created.

6. Missing Entitlement Import

Users can request access only after entitlements are imported.

Without Entitlement Import:

- Access catalog is empty.
 - Users cannot request permissions.
-

7. Provisioning Enabled Before Testing

Never enable provisioning before verifying:

- Connection
 - Authentication
 - Endpoint
 - Import jobs
-

8. Incorrect Feature Selection

Enabling unnecessary features increases complexity.

Enable only required features.

Troubleshooting

Always troubleshoot systematically.

Never make random configuration changes.

Troubleshooting Security System Issues

Symptoms

- Application not visible.
 - Accounts missing.
 - Provisioning fails.
-

Possible Causes

- Security System not created.
 - Wrong Security System selected.
 - Duplicate Security Systems.
-

Resolution

Verify:

- Security System exists.
 - Correct application mapping.
 - Endpoint association.
 - Connector assignment.
-

Troubleshooting Endpoint Issues

Symptoms

- Connection test fails.
 - Import jobs fail.
 - Provisioning fails.
-

Possible Causes

- Wrong URL.
 - Wrong credentials.
 - SSL certificate issue.
 - Firewall.
 - Network issue.
-

Resolution

Check:

- Target URL
 - Username
 - Password
 - SSL Certificate
 - Network connectivity
 - API availability
-

Troubleshooting Connector Issues

Symptoms

- Account Import fails.
 - Provisioning fails.
 - Reconciliation fails.
-

Possible Causes

- Connector configuration error.
 - API timeout.
 - Connector version mismatch.
 - Missing permissions.
 - Unsupported API changes.
-

Resolution

Verify:

- Connector configuration
 - Connector logs
 - Authentication
 - API permissions
 - Connector version compatibility
-

Troubleshooting Correlation Issues

Symptoms

- Duplicate identities.
 - Orphan accounts.
 - Accounts not linked.
 - Wrong user mapped.
-

Common Causes

- Username mismatch.
 - Employee ID mismatch.
 - Email mismatch.
 - Correlation rule incorrect.
-

Resolution

Check:

- Username mapping
 - Employee ID mapping
 - Email mapping
 - Correlation rules
 - Identity Repository
-

General Troubleshooting Workflow

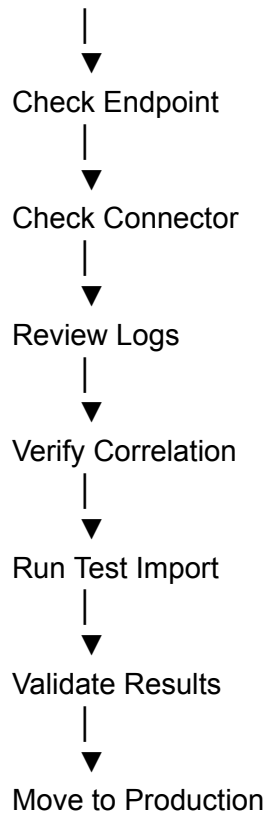
Problem Reported



Identify Affected Application



Check Connection



Real Project Troubleshooting Example

ABC Bank reports that new Salesforce accounts are not being created.

Investigation

Step 1

Connection Test

Result:

Success

Step 2

Account Import

Completed successfully.

Step 3

Provisioning Job

Failed.

Step 4

Connector Logs

Error:

Insufficient API permissions.

Step 5

Salesforce service account updated with required permissions.

Step 6

Provisioning rerun.

Accounts created successfully.

Interview Questions

Q1. What is the Saviynt Data Model?

The Saviynt Data Model defines how Users, Accounts, Security Systems, Endpoints, Connectors, and Entitlements are related to manage identity governance.

Q2. What is a Security System?

A Security System is the logical representation of an application inside Saviynt.

Q3. What is an Endpoint?

An Endpoint is the technical connection that stores the URL, authentication, credentials, and connector details required to communicate with an application.

Q4. What is a Connector?

A Connector is the communication component that enables Saviynt to import data, provision accounts, and reconcile information with target applications.

Q5. Why should each application have one Security System?

One Security System provides centralized governance and avoids duplicate configurations and reporting issues.

Q6. Why should Development and Production use separate Endpoints?

Separate Endpoints isolate environments, reduce risk, and allow testing without affecting production.

Q7. Why are OOTB Connectors preferred?

They are pre-built, tested, supported by Saviynt, and require less maintenance than custom connectors.

Q8. What is Identity Correlation?

Identity Correlation links imported application accounts to the correct Saviynt identity using attributes such as Employee ID, Username, or Email.

Q9. What causes orphan accounts?

Orphan accounts occur when imported accounts cannot be correlated with any identity due to incorrect or missing correlation attributes.

Q10. Why is Entitlement Import important?

It imports available permissions into Saviynt so users can request access and administrators can govern those permissions.

Q11. Why should connections be tested before onboarding?

Connection testing verifies connectivity, authentication, SSL configuration, and API availability before production jobs are enabled.

Q12. What information does a Connection contain?

A Connection stores the Target URL, authentication method, credentials, connector information, and communication settings.

Q13. What are common causes of Connection failures?

- Incorrect URL

- Invalid credentials
 - Expired passwords
 - SSL certificate issues
 - Firewall restrictions
 - Network outages
-

Q14. What are common Connector issues?

- API timeouts
 - Incorrect configuration
 - Missing permissions
 - Connector version mismatch
 - Unsupported API changes
-

Q15. What happens if Correlation fails?

Saviynt cannot link application accounts to identities, resulting in orphan accounts, duplicate identities, and inaccurate governance.

Q16. What is the difference between Account Import and User Import?

User Import creates identities from the Authoritative Source, while Account Import imports application-specific accounts from target systems.

Q17. Why is Reconciliation important?

Reconciliation synchronizes Saviynt with the latest account and entitlement information from target applications and detects changes made outside Saviynt.

Q18. Why should service accounts be used instead of personal accounts?

Service accounts provide better security, stability, auditing, and prevent integrations from breaking when employees leave.

Q19. What are the benefits of consistent naming conventions?

Consistent naming improves administration, troubleshooting, reporting, and maintenance.

Q20. What is the first step in troubleshooting provisioning failures?

Verify the Connection and perform a successful Connection Test before checking the Connector, Endpoint, logs, and target application.

Q21. Why should changes be tested in lower environments?

Testing in Development, Test, or UAT reduces the risk of production outages and validates configurations before deployment.

Q22. How do you troubleshoot Correlation issues?

Review correlation rules, compare usernames, employee IDs, and email addresses, inspect the Identity Repository, and rerun the correlation process after corrections.

Q23. What is the role of the Endpoint in troubleshooting?

The Endpoint is checked to verify URLs, credentials, authentication settings, SSL configuration, and network connectivity.

Q24. What is the role of the Connector in troubleshooting?

The Connector is checked to verify API communication, import jobs, provisioning logic, reconciliation, and connector logs.

Q25. Explain the complete Connected Application workflow.

1. Discover the application.
 2. Onboard the application.
 3. Create the Security System.
 4. Create the Endpoint.
 5. Configure the Connection.
 6. Test the Connection.
 7. Import Accounts.
 8. Import Entitlements.
 9. Correlate Accounts with Identities.
 10. Enable Provisioning.
 11. Run Reconciliation.
 12. Govern user access through requests, certifications, and audits.
-

Quick Revision

- **Security System** = Logical representation of an application.
- **Endpoint** = Technical connection to an application.
- **Connector** = Communication bridge between Saviynt and the application.
- **Connection** = Stores URL, credentials, and authentication details.
- **User Import** creates identities.
- **Account Import** imports application accounts.
- **Correlation** links accounts to identities.
- **Entitlement Import** imports available permissions.

- **Provisioning** sends changes from Saviynt to the application.
- **Reconciliation** synchronizes application data back into Saviynt.
- Use **OOTB Connectors**, separate Endpoints for each environment, dedicated service accounts, and regular reconciliation for reliable governance.
- Troubleshoot in a structured order: **Connection** → **Endpoint** → **Connector** → **Logs** → **Correlation** → **Import/Provisioning Jobs**.

Module 4 – Disconnected Application Onboarding

What is a Disconnected Application?

Definition

A **Disconnected Application** is an application that **cannot communicate directly with Saviynt** because it does not support connectors, APIs, or automated integration.

Instead of automatic provisioning and reconciliation, administrators use **manual processes, CSV files, databases, or ITSM tools like ServiceNow** to manage user access.

Simple Definition

A **Disconnected Application** is an application that cannot be integrated directly with Saviynt, so access management is performed manually or through file-based processes.

Why are Disconnected Applications Used?

Many organizations still use:

- Legacy Applications
- Homegrown Applications
- Custom ERP Systems
- Vendor Applications
- Older Applications without APIs

These applications may not support:

- REST APIs
- SOAP APIs
- LDAP
- Database connectivity

Instead of replacing these systems, organizations integrate them manually with Saviynt.

Characteristics of Disconnected Applications

- No direct connector
 - Manual provisioning
 - Manual deprovisioning
 - CSV/Excel-based imports
 - ITSM ticket integration
 - Manual verification
 - Administrator involvement
-

Connected vs Disconnected Applications

Connected	Disconnected
Direct connector available	No direct connector
Automatic provisioning	Manual provisioning
Automatic reconciliation	Manual reconciliation or file import
Real-time communication	Manual or scheduled updates
Minimal administrator effort	More administrator involvement

Import Data

Definition

Import Data is the process of bringing account or entitlement information from the disconnected application into Saviynt.

Since there is no connector, administrators use alternative methods.

Import Methods

- CSV Import
 - SOAP Import
 - REST Import
 - Database Import
 - Manual Upload
-

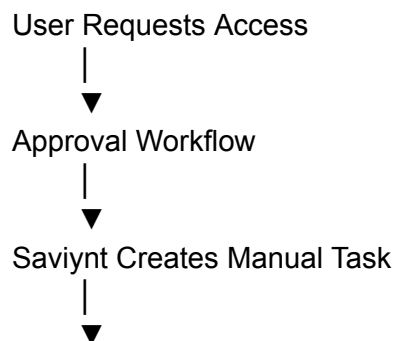
Manual Provisioning

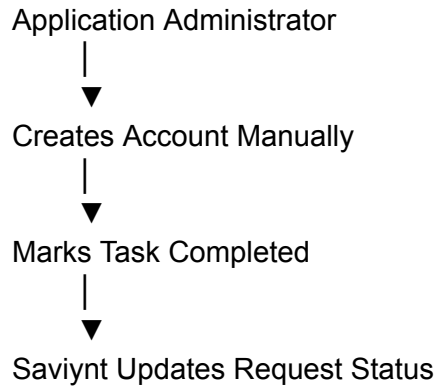
Definition

Manual Provisioning means the administrator creates, modifies, or removes user access manually after receiving a request from Saviynt.

Saviynt records the approval and tracks the task, but the actual work is performed by an administrator.

Workflow





CSV Import

Definition

CSV Import uses comma-separated files to import accounts or entitlements into Saviynt.

Example

Username,Department,Role
rahul,Finance,Finance_User
priya,HR,HR_Manager

Advantages

- Easy to create
 - Supported by almost every application
 - Good for bulk imports
-

Limitations

- Manual preparation
- Not real-time

- Human errors possible
-

SOAP Import

Definition

Some legacy enterprise applications expose SOAP Web Services instead of REST APIs.

Saviynt can consume SOAP services to retrieve user or entitlement information.

Characteristics

- XML-based communication
 - Secure enterprise integrations
 - Common in older ERP systems
-

REST Import

Definition

Some applications provide REST APIs for importing account information even if they do not support complete provisioning.

Saviynt retrieves data using:

- HTTP
 - HTTPS
 - JSON
-

Example

Saviynt sends:

GET /users

Application returns:

```
[  
  {  
    "username":"rahul",  
    "department":"Finance"  
  }  
]
```

Database Import

Definition

Saviynt imports user and account information directly from a database.

Supported databases include:

- Oracle
 - SQL Server
 - PostgreSQL
 - MySQL
-

Example

SQL Query

```
SELECT username, department  
FROM employee_accounts;
```

Custom Application

Definition

A Custom Application is an application that has no standard connector.

Administrators configure:

- Manual imports
 - Custom mappings
 - Manual provisioning
 - Custom workflows
-

Example

An internally developed Leave Management System.

No APIs.

CSV files are exchanged daily.

Provisioning Methods

Disconnected applications support different provisioning methods.

Manual Provisioning

Administrator performs all actions manually.

CSV-Based Provisioning

Saviynt generates a CSV.

Administrator imports the file into the application.

Ticket-Based Provisioning

Saviynt creates a ServiceNow ticket.

Administrator performs provisioning.

Ticket is updated after completion.

Database-Based Provisioning

Administrator executes SQL scripts or database procedures.

Manual Tasks

Definition

Manual Tasks are work items created by Saviynt when automation is not available.

Examples

- Create user account
 - Reset password
 - Assign role
 - Remove access
 - Disable account
-

Workflow

Request Approved

↓

Manual Task Created

↓

Administrator Completes Task

↓

Task Closed

↓

Saviynt Updated

ServiceNow Integration

Why Use ServiceNow?

Many organizations already manage IT requests through ServiceNow.

Saviynt integrates with ServiceNow to automate task creation.

Workflow

Access Request

↓

Approval

↓

ServiceNow Ticket Created

↓

Administrator Completes Task

↓

Ticket Closed



Saviynt Updated

Benefits

- Existing ITSM process reused
 - Complete audit trail
 - Better tracking
 - Reduced email communication
-

Enrollment Steps

Typical onboarding steps for a disconnected application:

Step 1

Create Application.

Step 2

Create Security System.

Step 3

Create Endpoint.

Step 4

Configure manual provisioning.

Step 5

Configure account import.

Step 6

Configure entitlement import.

Step 7

Configure approval workflow.

Step 8

Test import process.

Step 9

Move to production.

Upload Accounts

Accounts are imported using:

- CSV
- Database
- REST
- SOAP

Each imported account is later correlated with identities.

Upload Entitlements

Entitlements are imported manually.

Examples:

Groups

Roles

Permissions

Profiles

These become requestable inside Saviynt.

Complete Workflow

Disconnected Application



CSV / REST / SOAP / Database



Import Accounts



Import Entitlements



Identity Correlation



Access Request

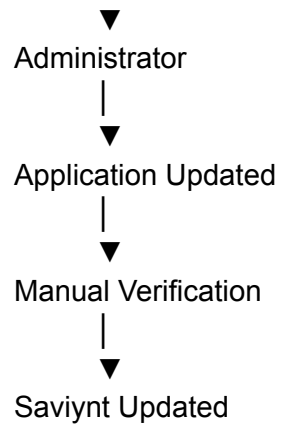


Approval



Manual Provisioning Task





Real Project Example

ABC Manufacturing has an old ERP application.

The application:

- Has no API
- No connector
- No LDAP

Step 1

Administrator creates a Security System.

Step 2

Creates an Endpoint.

Step 3

Uploads employee accounts using CSV.

Step 4

Imports application roles.

Step 5

Employees request access through Saviynt.

Step 6

Manager approves.

Step 7

Saviynt creates a ServiceNow ticket.

Step 8

ERP Administrator manually creates the account.

Step 9

Administrator closes the ServiceNow ticket.

Step 10

Saviynt updates the request status and records the activity for audit.

Advantages

- Supports legacy systems
- No application replacement required

- Lower implementation cost
 - Integrates with manual business processes
 - Provides governance even without automation
-

Limitations

- Manual effort required
 - Slower provisioning
 - Human error possible
 - No real-time synchronization
 - Higher operational overhead
-

Best Practices

- Prefer connected applications whenever possible.
 - Use standardized CSV templates.
 - Validate imported data before processing.
 - Configure approval workflows for manual tasks.
 - Integrate with ServiceNow for better tracking.
 - Maintain accurate account-to-user correlation.
 - Periodically review manually provisioned accounts.
 - Document manual provisioning procedures.
 - Test imports in lower environments before production.
-

Interview Questions

Q1. What is a Disconnected Application?

A Disconnected Application is an application that cannot communicate directly with Saviynt through a connector, so access is managed manually or through file-based imports.

Q2. Why are Disconnected Applications used?

They are used for legacy, homegrown, or custom applications that do not support direct integration methods such as REST, SOAP, or LDAP.

Q3. What are common import methods for Disconnected Applications?

CSV Import, SOAP Import, REST Import, Database Import, and Manual Upload.

Q4. What is Manual Provisioning?

Manual Provisioning is the process where an administrator manually creates, updates, or removes user access after Saviynt generates and approves the request.

Q5. Why is ServiceNow commonly integrated with Disconnected Applications?

ServiceNow manages manual provisioning tasks, provides auditability, tracks task status, and integrates with existing IT service management processes.

Q6. What is the difference between Connected and Disconnected Applications?

Connected Applications support automated provisioning and reconciliation through connectors, while Disconnected Applications rely on manual processes or file-based imports.

Q7. What are the limitations of Disconnected Applications?

They require manual effort, are not real-time, are more prone to human error, and have higher operational overhead.

Quick Revision

- **Disconnected Applications** do not support direct integration with Saviynt.
- Account and entitlement data is imported using **CSV, REST, SOAP, Database, or Manual Uploads**.
- Provisioning is typically **manual** or **ticket-based**.
- **ServiceNow** is commonly used to manage manual provisioning tasks.
- Imported accounts are correlated with identities before governance begins.
- Manual processes still provide governance, approvals, and audit trails even without full automation.
- Disconnected onboarding is commonly used for **legacy, homegrown, and custom applications**.

Module 5 – Integration Agent for Disconnected Applications

What is Saviynt Integration Agent?

Definition

The **Saviynt Integration Agent (SIA)** is a lightweight software component that enables Saviynt to communicate with systems that cannot be directly integrated using standard connectors.

It acts as a secure bridge between Saviynt Identity Cloud and applications inside an organization's network or applications requiring special automation.

Unlike traditional disconnected applications that require manual work, the Integration Agent can automate many of these tasks.

Simple Definition

Saviynt Integration Agent is a secure bridge that allows Saviynt to automate tasks on systems that cannot be accessed directly from the cloud.

Why is the Integration Agent Needed?

Many organizations have applications that:

- Are behind a firewall.
- Do not expose APIs.
- Support only browser-based administration.
- Are legacy systems.
- Cannot be connected directly to Saviynt Cloud.

Without the Integration Agent:

- Provisioning remains manual.
- Account management is slower.
- Human errors increase.

The Integration Agent solves these challenges by securely executing tasks inside the customer environment.

Benefits of the Integration Agent

- Automates disconnected application management.
 - Reduces manual provisioning effort.
 - Supports legacy applications.
 - Works securely behind firewalls.
 - Improves consistency.
 - Reduces human errors.
 - Provides audit trails.
 - Supports browser automation.
 - Integrates AI capabilities.
-

AI Agent

Definition

An **AI Agent** is an intelligent software component capable of understanding instructions, making decisions within defined rules, and performing tasks automatically.

In Saviynt, AI Agents help automate repetitive identity management tasks.

Examples

- Creating accounts
- Assigning roles
- Filling web forms

- Updating user information
 - Validating provisioning steps
-

CUA (Computer Use Agent)

Definition

CUA (Computer Use Agent) is an AI-powered automation component that interacts with applications through their graphical user interface (GUI), similar to how a human administrator would.

Instead of calling APIs, it performs actions such as:

- Opening web pages
 - Clicking buttons
 - Entering usernames
 - Selecting roles
 - Saving changes
-

Why is CUA Important?

Some legacy applications have:

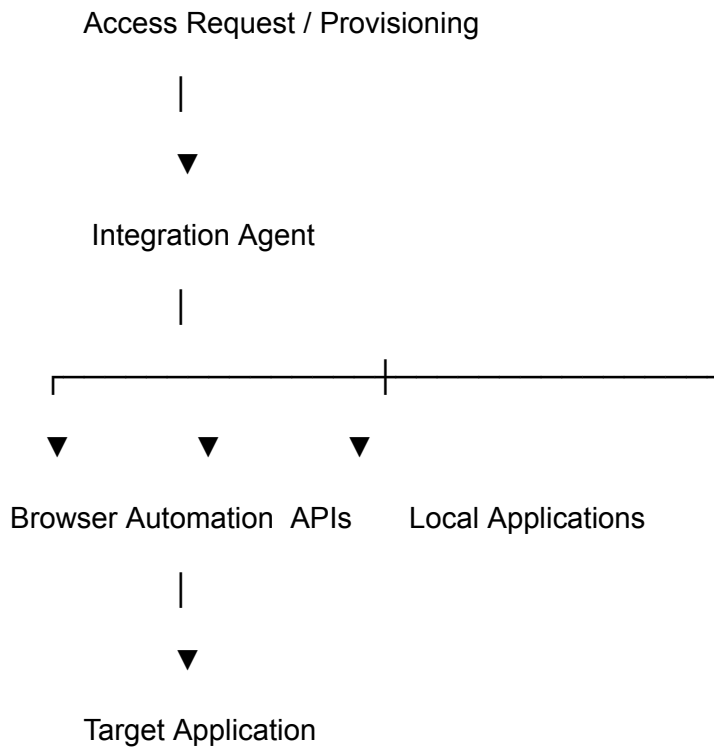
- No APIs
- No connectors
- No database access

CUA enables automation by interacting with the application's user interface.

Integration Agent Architecture

Saviynt Cloud

|



The Integration Agent receives tasks from Saviynt and executes them using the appropriate method.

Authentication Flow

Before performing any task, the Integration Agent must authenticate.

Workflow



|



Target Application

|

Authentication Successful

Authentication methods may include:

- Username and Password
- API Token
- OAuth
- Certificates

Provisioning Flow

Provisioning through the Integration Agent follows these steps:

Access Request

|



Approval

|



Saviynt Creates Provisioning Task

|



Integration Agent Receives Task

|



Agent Executes Task



Application Updated



Status Returned to Saviynt

Validation

After performing a task, the Integration Agent validates whether the action was successful.

Examples:

- Was the account created?
- Was the role assigned?
- Was the password reset?
- Was the account disabled?

Validation ensures accurate execution and reduces provisioning errors.

Guardrails

Definition

Guardrails are predefined security rules that limit what the AI Agent or Integration Agent is allowed to do.

Examples

- Only approved provisioning tasks may be executed.
 - Only authorized applications may be accessed.
 - Sensitive operations require additional approval.
 - Prevent deletion of critical accounts.
 - Restrict administrative actions.
-

Why Guardrails are Important?

They prevent:

- Unauthorized automation
 - Accidental changes
 - Security violations
 - Compliance issues
-

Task Processing

Every provisioning request becomes a task.

The Integration Agent continuously checks for pending tasks.

Workflow

Provisioning Request

|



Task Queue

|



Integration Agent

|



Execute Task



Update Status

Task Status

Each task has a lifecycle.

Common statuses include:

- Pending
- In Progress
- Completed
- Failed
- Retry
- Cancelled

Administrators can monitor task progress directly from Saviynt.

Browser Automation

Definition

Browser Automation allows the Integration Agent to interact with web applications exactly as a human administrator would.

Typical actions include:

- Open browser
- Login

- Navigate menus
 - Fill forms
 - Click buttons
 - Submit changes
 - Logout
-

Example

Legacy HR Application

↓

No API

↓

Integration Agent opens browser

↓

Logs in

↓

Creates user account

↓

Logs out

LLM (Large Language Model)

Definition

A **Large Language Model (LLM)** is an AI model capable of understanding natural language instructions and assisting with intelligent task execution.

In the context of Saviynt, an LLM can help:

- Interpret automation instructions.

- Assist browser automation.
- Understand UI changes.
- Generate responses within predefined guardrails.

The LLM **does not make unrestricted security decisions**; it operates within approved workflows and governance controls.

Processing Queue

Definition

The Processing Queue stores all pending tasks waiting to be executed.

Tasks are processed one by one or in batches, depending on configuration.

Benefits:

- Organized processing
 - Retry support
 - Better scalability
 - Load balancing
-

Logs

Every action performed by the Integration Agent is logged.

Typical log information:

- Task ID
- User
- Application
- Timestamp
- Action Performed
- Success/Failure
- Error Message

Logs are essential for:

- Troubleshooting
 - Auditing
 - Compliance
-

Data Privacy

Saviynt is designed to minimize the exposure of customer data.

Best practices include:

- Encrypt communication.
 - Store only required information.
 - Limit access to authorized users.
 - Mask sensitive information where appropriate.
-

Security

Security features include:

- Secure communication (HTTPS/TLS)
 - Authentication
 - Authorization
 - Encryption
 - Audit logging
 - Role-based access control
 - Least privilege service accounts
-

Customer Data Protection

Saviynt protects customer information by:

- Encrypting data in transit.
- Encrypting data at rest (where applicable).
- Using secure service accounts.
- Maintaining detailed audit logs.

- Restricting access based on roles.
- Following organizational security policies and compliance requirements.

Complete Integration Agent Workflow

User Requests Access



Approval



Provisioning Task Created



Processing Queue



Integration Agent



Authentication



Browser/API Automation





Validation



Update Task Status



Saviynt Audit Logs Updated

Real Project Example 1 – Legacy ERP System

ABC Manufacturing uses a legacy ERP application.

The ERP:

- Has no API.
- Has no connector.
- Supports only a web interface.

Solution

1. User requests ERP access in Saviynt.
2. Manager approves.
3. Integration Agent receives the provisioning task.
4. CUA opens the ERP web portal.
5. Logs in with a service account.
6. Creates the user account.
7. Assigns the required role.
8. Validates the account creation.
9. Reports success back to Saviynt.

No administrator performs the steps manually.

Real Project Example 2 – Internal HR Portal

A company has an internal HR application with no connector.

The Integration Agent:

- Logs into the HR portal.
 - Navigates to the user management page.
 - Creates employee accounts.
 - Assigns department permissions.
 - Logs the operation.
 - Updates Saviynt with the completion status.
-

Advantages

- Automates disconnected applications.
 - Supports AI-driven automation.
 - Reduces manual effort.
 - Improves provisioning speed.
 - Maintains auditability.
 - Works with legacy systems.
 - Enhances operational efficiency.
-

Limitations

- Browser automation may require maintenance if the application's UI changes.
 - Requires secure configuration and monitoring.
 - Performance depends on the target application's responsiveness.
 - Guardrails must be carefully designed to prevent unintended actions.
-

Best Practices

- Use the Integration Agent only when direct connectors are not available.
 - Configure dedicated service accounts.
 - Test automation in non-production environments.
 - Enable detailed logging.
 - Review failed tasks regularly.
 - Keep browser automation scripts updated.
 - Apply guardrails to sensitive operations.
 - Protect credentials using secure storage.
 - Monitor the processing queue and task status.
-

Interview Questions

Q1. What is the Saviynt Integration Agent?

The Saviynt Integration Agent is a secure software component that enables Saviynt to automate identity management tasks for applications that cannot be integrated directly through standard connectors.

Q2. Why is the Integration Agent needed?

It provides secure communication and automation for legacy, on-premises, or disconnected applications that do not support APIs or direct connectors.

Q3. What is a Computer Use Agent (CUA)?

A Computer Use Agent is an AI-powered automation component that performs actions on an application's user interface, such as clicking buttons and filling forms, similar to a human administrator.

Q4. What is Browser Automation?

Browser Automation allows the Integration Agent to interact with web applications through their graphical user interface when APIs are unavailable.

Q5. What are Guardrails?

Guardrails are predefined security controls that restrict what automated agents can do, ensuring actions remain within approved governance policies.

Q6. What is the purpose of Validation?

Validation confirms that provisioning tasks were completed successfully before updating Saviynt.

Q7. What information is stored in Integration Agent logs?

Logs typically include task IDs, user details, application names, timestamps, actions performed, success or failure status, and error messages.

Q8. How does the Processing Queue work?

Provisioning tasks are placed into a queue, processed by the Integration Agent, executed in the target application, and then updated with the final status.

Q9. How does the Integration Agent protect customer data?

It uses secure authentication, encrypted communication, role-based access control, service accounts, audit logging, and follows organizational security policies to protect customer data.

Q10. When should you use the Integration Agent?

Use it when an application cannot be integrated directly through standard Saviynt connectors or APIs, but automation is still required.

Quick Revision

- **Integration Agent** securely connects Saviynt with disconnected or on-premises applications.
- **AI Agent** automates identity-related tasks within approved governance rules.
- **CUA (Computer Use Agent)** performs browser-based automation for applications without APIs.
- **Guardrails** enforce security restrictions on automated actions.
- **Validation** confirms provisioning success.
- **Processing Queue** manages pending automation tasks.
- **Logs** provide complete audit and troubleshooting information.
- The Integration Agent improves automation while maintaining security, compliance, and governance for disconnected applications.

Module 6 – Best Practices & Troubleshooting

Introduction

After onboarding applications, administrators must ensure that the environment remains secure, stable, and easy to maintain. Following best practices reduces implementation errors, improves performance, simplifies troubleshooting, and helps meet compliance requirements.

Best Practices

1. Security System Naming

What is Security System Naming?

The Security System name should clearly represent the application being managed.

Good Examples

- ActiveDirectory
- Salesforce
- AWS
- SAP
- Workday

Bad Examples

- App1
- Test123
- ABC

Why is Naming Important?

Proper naming:

- Makes administration easier.
 - Improves reporting.
 - Simplifies troubleshooting.
 - Avoids duplicate Security Systems.
 - Helps new administrators understand the environment.
-

Best Practice

Use the application name as the Security System name.

One Security System should normally represent one application.

2. Endpoint Naming

What is Endpoint Naming?

Endpoints should clearly identify the environment.

Good Examples

- Salesforce_PROD
- Salesforce_DEV
- AWS_UAT
- SAP_QA

Bad Examples

- Endpoint1
 - Server2
 - Test
-

Why?

Organizations usually have multiple environments.

Proper names prevent accidental provisioning to the wrong environment.

3. Standard Workflows

What are Standard Workflows?

Standard Workflows define the approval process for:

- Access Requests
 - Provisioning
 - Deprovisioning
 - Certifications
-

Why Use Standard Workflows?

Benefits include:

- Consistent approvals
 - Easier maintenance
 - Faster troubleshooting
 - Compliance
 - Reduced customization
-

Example Workflow

User Request

|



Manager Approval

|



Application Owner Approval

|



Provisioning

4. Identity Access Types

Definition

Identity Access Types define how access is assigned.

Common types include:

- Birthright Access
- Requestable Access
- Role-Based Access
- Temporary Access
- Emergency Access

Birthright Access

Automatically assigned.

Example:

Every employee receives:

- Email
 - Active Directory
-

Requestable Access

Requires user request and approval.

Example:

SAP Finance Role

Temporary Access

Access expires automatically.

Example:

Database Admin access for 7 days.

Emergency Access

Granted quickly for production issues.

Automatically revoked after the emergency.

5. Configuration Settings

Before enabling production, verify:

- Connection
 - Authentication
 - Endpoint
 - Security System
 - Connector
 - Import Jobs
 - Provisioning Jobs
 - Reconciliation Jobs
-

Best Practice

Never enable provisioning before validating every configuration.

6. Why Avoid Special Characters?

Avoid using:

- Spaces
- %
- &
- @
- !

○

in object names.

Why?

Special characters may cause:

- Import failures
 - Script issues
 - SQL problems
 - API errors
 - Search difficulties
-

Best Practice

Use:

Salesforce_PROD

instead of:

7. Why Use Reusable Workflows?

Instead of creating multiple similar workflows,
reuse existing workflows.

Benefits

- Less maintenance.
 - Easier testing.
 - Consistent approvals.
 - Reduced errors.
-

8. Entitlement Type Best Practices

Keep Entitlement Types broad.

Examples:

- Group
 - Role
 - Permission
 - Profile
-

Why?

One Entitlement Type can contain hundreds of Entitlement Values.

This keeps the data organized.

9. Certifiable Option

What is Certifiable?

The **Certifiable** option determines whether an entitlement should be included in Access Certification campaigns.

When Enable?

Enable for:

- Business Roles
 - Privileged Roles
 - Sensitive Groups
-

When Disable?

Disable for:

- Technical permissions
 - Temporary objects
 - Internal system values
-

10. Request Option

Definition

The **Request Option** determines whether users can request an entitlement.

Enable

If users should request access.

Example:

Finance Group

Disable

If access should only be assigned automatically.

Example:

System accounts

11. Required Option

Definition

The **Required Option** marks an entitlement as mandatory.

Example

Every employee must have:

- Email
- Active Directory

These are marked Required.

Users cannot remove them.

12. Order Index

Definition

Order Index controls how entitlements appear in the user interface.

Why Use It?

Provides:

- Better user experience.
 - Logical ordering.
 - Easier navigation.
-

Troubleshooting

Always troubleshoot systematically.

Do not change multiple configurations simultaneously.

1. Application Discovery Issues

Symptoms

Application not discovered.

Possible Causes

- Incorrect Discovery Source.
- CMDB unavailable.

- CSV format incorrect.
 - Missing permissions.
-

Resolution

Check:

- Discovery configuration.
 - ServiceNow connection.
 - CSV format.
 - Logs.
-

2. Connection Failures

Symptoms

Connection Test fails.

Causes

- Wrong URL.
 - Invalid credentials.
 - Firewall.
 - SSL issue.
-

Resolution

Verify:

- URL
- Username
- Password
- Certificates

- Network connectivity
-

3. Authentication Failures

Symptoms

Unable to log in.

Causes

- Incorrect password.
 - Expired password.
 - Disabled account.
 - OAuth issue.
-

Resolution

Update credentials and retest authentication.

4. Connector Failures

Symptoms

Account Import fails.

Provisioning fails.

Causes

- Connector misconfiguration.
 - API changes.
 - Missing permissions.
 - Unsupported version.
-

Resolution

Check:

- Connector logs.
 - API documentation.
 - Connector configuration.
 - Service account permissions.
-

5. Endpoint Errors

Symptoms

Endpoint unavailable.

Causes

- Wrong endpoint.
 - Invalid environment.
 - Incorrect URL.
-

Resolution

Verify endpoint configuration and environment mapping.

6. Security System Errors

Symptoms

Duplicate applications.

Missing accounts.

Provisioning issues.

Causes

- Duplicate Security Systems.
 - Incorrect mappings.
 - Missing Endpoint.
-

Resolution

Verify:

- Security System.
 - Endpoint relationship.
 - Connector assignment.
-

7. Mapping Errors

Symptoms

Duplicate users.

Wrong provisioning.

Missing accounts.

Causes

- Username mismatch.
- Employee ID mismatch.
- Manager mismatch.
- Department mismatch.

Resolution

Review attribute mapping and correlation rules.

8. Import Failures

Symptoms

Accounts not imported.

Causes

- CSV errors.
 - API unavailable.
 - Network issue.
 - Invalid mappings.
-

Resolution

Check:

- Import logs.
- CSV file.

- API response.
 - Mapping configuration.
-

9. Provisioning Failures

Symptoms

Account not created.

Role not assigned.

Causes

- Provisioning job failed.
 - Authentication issue.
 - API failure.
 - Permission issue.
-

Resolution

Check:

- Provisioning logs.
 - Connector logs.
 - Application logs.
 - Service account permissions.
-

10. Reconciliation Failures

Symptoms

Latest application changes not visible in Saviynt.

Causes

- Reconciliation job failed.
 - Connector issue.
 - Network issue.
 - Incorrect mappings.
-

Resolution

Verify:

- Reconciliation job.
 - Connector.
 - Endpoint.
 - Logs.
-

11. Marketplace Issues

Symptoms

Connector unavailable.

Marketplace inaccessible.

Causes

- Internet issue.
 - Version mismatch.
 - Unsupported connector.
-

Resolution

Verify Marketplace connectivity and compatibility.

12. ServiceNow Discovery Issues

Symptoms

Applications not discovered.

Causes

- CMDB unavailable.
 - Wrong credentials.
 - Missing permissions.
-

Resolution

Check:

- CMDB configuration.
 - Service account.
 - Discovery logs.
-

13. CSV Upload Issues

Common Errors

- Missing headers.
- Wrong delimiter.
- Duplicate usernames.

- Empty mandatory fields.
 - Wrong encoding.
-

Resolution

Validate CSV before uploading.

Use UTF-8 encoding.

14. Integration Agent Issues

Symptoms

Automation not executed.

Tasks remain pending.

Causes

- Agent offline.
 - Authentication issue.
 - Browser automation failure.
 - Guardrail restriction.
-

Resolution

Check:

- Agent status.
- Task queue.
- Logs.
- Browser configuration.
- Credentials.

Complete Troubleshooting Workflow

Problem Reported



Identify Application



Check Security System



Check Endpoint



Check Connection



Check Authentication



Check Connector



Review Logs

|



Run Import / Provisioning Test

|



Validate Results

|



Move to Production

Real Project Example

ABC Bank onboarded Salesforce.

Users reported that new accounts were not being created.

Investigation

1. Connection Test – Successful.
2. Authentication – Successful.
3. Endpoint – Correct.
4. Provisioning Job – Failed.
5. Connector Logs – "Insufficient API Permission."
6. Salesforce service account permissions were updated.
7. Provisioning Job rerun.
8. Accounts created successfully.

Root Cause: Service account lacked the required API permissions.

Interview Questions

Q1. Why are naming conventions important?

They improve administration, reporting, troubleshooting, and prevent duplicate configurations.

Q2. Why should special characters be avoided?

They can cause issues with APIs, scripts, SQL queries, and integrations.

Q3. What is the purpose of the Certifiable option?

It determines whether an entitlement appears in Access Certification campaigns.

Q4. What does the Request Option control?

It determines whether users can request a particular entitlement through Saviynt.

Q5. What is the Required option?

It marks mandatory access that users should always have and cannot remove.

Q6. Why use reusable workflows?

Reusable workflows reduce duplication, simplify maintenance, and provide consistent approval processes.

Q7. What are common provisioning failures?

Authentication issues, connector failures, permission problems, API errors, and incorrect mappings.

Q8. How do you troubleshoot a failed Connection Test?

Check the Target URL, credentials, SSL certificates, firewall, network connectivity, and authentication settings.

Q9. What are common CSV upload problems?

Missing headers, incorrect delimiters, duplicate usernames, empty mandatory fields, and wrong file encoding.

Q10. How do you troubleshoot Integration Agent issues?

Verify the agent is running, review the task queue and logs, confirm authentication, validate browser automation, and check guardrail restrictions.

Quick Revision

- Use clear **Security System** and **Endpoint** naming conventions.
- Follow **standard and reusable workflows** wherever possible.
- Use **Birthright**, **Requestable**, **Temporary**, and **Emergency** access appropriately.
- Avoid **special characters** in object names.
- Configure **Certifiable**, **Request**, **Required**, and **Order Index** correctly for entitlements.
- Troubleshoot systematically: **Security System** → **Endpoint** → **Connection** → **Authentication** → **Connector** → **Logs** → **Import/Provisioning/Reconciliation Jobs**.
- Validate configurations in lower environments before production deployment.
- Keep logs enabled and monitor Integration Agent health regularly.